

Phospholipid Scramblase 1 (PLSCR1) Regulates Interferon-Lambda Receptor 1 (IFN- λ R1) and IFN- λ Signaling in Influenza A Virus (IAV) Infection

Reviewed Preprint

v2 • November 21, 2025

Revised by authors

Reviewed Preprint

v1 • March 11, 2025

Alina X Yang, Lisa Ramos-Rodriguez, Parand Sorkhdini, Dongqin Yang, Carmelissa Norbrun, Sonoor Majid, Sanghyun Lee, Yong Zhang, Michael J Holtzman, David F Boyd, Yang Zhou ✉

Department of Molecular Microbiology and Immunology, Brown University, Providence, United States • Division of Pulmonary and Critical Care Medicine, Washington University School of Medicine in St. Louis, St Louis, United States • Department of Molecular, Cell and Developmental Biology, University of California, Santa Cruz, Santa Cruz, United States

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This **valuable** manuscript presents a potentially novel mechanism by which the phospholipid scramblase, PLSCR1, defends against influenza A virus infection. The strength of the paper rests on **solid** findings involving knockout and lung specific over-expressing Plscr1 mice, airway tissue expression and mechanistic studies to show Plscr1 enhances type III interferon-mediated viral clearance.

<https://doi.org/10.7554/eLife.104359.2.sa2>

Abstract

Phospholipid scramblase 1 (PLSCR1) is an antiviral interferon-stimulated gene (ISG) that has several known anti-influenza functions. However, the mechanisms in relation to its expression compartment and enzymatic activity have not been completely explored. Moreover, only limited animal models have been studied to delineate its role at the tissue level in influenza infections. Our results showed that Plscr1 expression was highly induced by influenza A virus (IAV) infection *in vivo* and in airway epithelial cells treated with IFN- λ . We found that infected *Plscr1*^{-/-} mice exhibited exacerbated body weight loss, decreased survival rates, heightened viral replication, and increased lung damage. Interestingly, transcriptomic analyses demonstrated that Plscr1 was required for the expression of type 3 interferon receptor (Ifn- λ r1) and a large subset of ISGs upon IAV infection. The impaired expression of Ifn- λ r1 and downstream ISGs may be responsible for delayed viral clearance and worse lung inflammation in *Plscr1*^{-/-} mice. PLSCR1 acts as a transcriptional activator of *IFN- λ R1* by directly binding to its promotor after IAV infection. In addition, PLSCR1 interacted with IFN- λ R1 on the cell surface of pulmonary epithelial cells following IAV infection,

suggesting it also modulated IFN- λ signaling via protein-protein interactions. The lipid scramblase activity of PLSCR1 was found to be dispensable for its anti-flu activity. Finally, single-cell RNA sequencing data indicated that *Plscr1* expression was significantly upregulated in ciliated airway epithelial cells in mice following IAV infection. Consistently, *Plscr1^{flloxStop}Foxj1-Cre⁺* mice with ciliated epithelial cell-specific *Plscr1* overexpression showed reduced susceptibility to IAV infection, less inflammation and enhanced *Ifn- λ 1* expression, suggesting that *Plscr1* primarily regulates type 3 IFN signaling as a cell intrinsic defense factor against IAV in ciliated airway epithelial cells. Our research will elucidate virus-host interactions and pave the way for the development of novel anti-influenza drugs that target human elements like PLSCR1, thereby mitigating the emergence of drug-resistant IAV strains.

Introduction

Influenza A virus (IAV) is highly contagious and causes acute respiratory infectious disease transmitted by virus-containing droplets. Its segmented gene and wide host range facilitate frequent antigenic shift during coinfection, posing a significant threat to human health. IAV has caused several pandemics in human history, and seasonal flu remains a major health burden in present days, despite annual vaccination efforts[1]. Existing anti-flu drugs mainly target influenza virus proteins: 1) penetration and shelling inhibitors, such as Gocovri (amantadine) and Flumadine (Rimantadine); 2) neuraminidase inhibitors, such as Relenza (zanamivir), Tamiflu (oseltamivir phosphate), Rapivab (peramivir) and Xofluza (baloxavir marboxil)[2]. However, the emergence of drug-resistant variant strains is common. Moreover, these direct antivirals have a short therapeutic window and are the most effective only if given within the first 48 hours after the initial infection[2]. In addition, IAV-induced acute inflammation could persist, leading to severe complications such as life-threatening pneumonia, immunopathology, and acute respiratory distress syndrome (ARDS). Therefore, there is a pressing need to understand host immune responses and develop anti-influenza drugs targeting host factors.

At the center of anti-flu immunity are the interferon (IFN) pathways. Discovered a mere two decades ago, type 3 IFNs were initially perceived as redundant to type 1 IFNs, given their shared intracellular signaling pathways and antiviral activities. However, recent studies have revealed their unique properties, notably their signaling through a pair of heterodimer receptors (IFN- λ R1/IL-10R2) distinct from type 1 IFN receptor complex (IFN- α R1/IFN- α R2)[3, 4]. First of all, IFN- λ exhibits a more constrained expression pattern compared to type 1 IFNs. While the cellular sources of type 1 IFNs in viral infections depend on the infection route and the tissue tropism, they can be produced by a large variety of cell types, including epithelial, parenchymal, immune and stromal cells, to combat infections in the skin, mucosal, organ, lymph node, and at the systemic level[5]. In contrast, IFN- λ is primarily secreted by cells at barrier surfaces, such as respiratory and gastrointestinal epithelial cells, DCs and macrophages[6, 7]. Moreover, their receptor distributions are vastly different. IFN- α R1/IFN- α R2 complex is present on nearly all cell types, but IFN- λ R1/IL-10R2 complex is expressed exclusively on epithelial cells and few immune cells, including neutrophils and subsets of dendritic cells[8, 9]. Most importantly, IFN- λ is produced earlier than type 1 interferons in IAV infection and can elicit effective antiviral responses without inflammation such as the release of tissue-damaging mediators such as tumor necrosis factor (TNF) and IL-1 β from neutrophils. In fact, only under a high dose of IAV are type 1 interferons detected, contributing to tissue immunopathology [10]. Another exclusive anti-flu mechanism of IFN- λ is its role in preventing viral transmission from the upper airways to the lungs[11]. Taken together, IFN- λ provides a non-redundant front-line shield against influenza virus.

IFNs secreted by infected cells signal neighboring cells to enter an antiviral state by inducing the expression of hundreds of ISGs[12]. PLSCR1, the most studied member of the phospholipid scramblase protein family, is one such ISG, with its expression highly induced by type 1, 2 and 3

interferons in various viral infections[13–15]. It is a type II transmembrane protein generally located on the cell membrane, but can be imported into the nucleus and act as a transcriptional factor to regulate several gene expressions by directly binding to their promoter regions[16, 17]. PLSCR1 is detected in all tissues and has a relatively high expression in lung. Although it is universally expressed across various cell types in the lung including alveolar epithelial cells and lymphocytes, macrophages have the highest expression, followed by endothelial cells and respiratory ciliated cells[18]. The main function of PLSCR1 is to catalyze Ca^{2+} -dependent, ATP-independent, bidirectional and non-specific translocation of phospholipids between inner & outer leaflet of plasma membrane. Scrambling of membrane phospholipids results in the externalization of phosphatidylserine (PS), which acts as a docking site for many biological processes including coagulation, apoptosis, and activation[19].

Previous studies have described some critical anti-influenza activities of PLSCR1. For example, PLSCR1 interacts with the nucleoprotein (NP) of IAV, impairing its nuclear import and thereby suppressing virus replication in A549 cells[20]. In addition, Plscr1 competes with immunoglobulin-like domain-containing receptor 1 (ILDR1) for NP binding, inhibiting swine influenza virus (SIV) infection in mice[21]. Besides direct interactions with the virus, PLSCR1 interacts with toll-like receptor (TLR) 9 and regulates its trafficking and ability to induce type I interferon production in plasmacytoid dendritic cells[22]. Furthermore, PLSCR1 is required to potentiate the expression of numerous other ISGs in response to IFN- β in Hey1B cells, including p56, ISG15 and OAS[23]. The goals of this project are to determine the roles of Plscr1 in an IAV-infected mouse model, to implicate its involvement in IFN- λ signaling, and to elucidate the cell types responsible for Plscr1-mediated anti-influenza activities.

To address this knowledge gap, we have employed a systematic approach, leveraging both global and cell type-specific *Plscr1*^{-/-} and *Plscr1* knock-in overexpression mice infected with mouse-adapted human IAV. Human respiratory epithelial cell lines harboring PLSCR1 mutations have also been used to elucidate the subcellular roles of PLSCR1. Our findings demonstrate that Plscr1 is an IFN- λ -stimulated gene, and it protects mice against IAV Infection by regulating *IFN- λ R1* gene expression in the nucleus and interacting with IFN- λ R1 protein on the cell membrane in ciliated airway epithelial cells. Our research will aid in the understanding of virus-host interactions and the development of novel anti-influenza therapeutics targeting human components, preventing the emergence of drug-resistant IAV strains.

Results

Plscr1 Suppresses Viral Replication and Protects Mice in IAV Infection

To uncover any anti-influenza functions of Plscr1 in mice, we employed *Wt* and *Plscr1*^{-/-} mice and exposed them to IAV infection. We pursued a systematic approach including multiple endpoints and parameters to monitor mouse health, viral replication and dissemination, antiviral and inflammatory responses, and tissue damage (Figure 1A). We established 300 plaque-forming unit (pfu) of IAV as the sublethal dose, and 900 pfu as the lethal dose resulting in ~50% mortality in *Wt* mice (LD50). Significantly higher *Plscr1* expression was observed in *Wt* infected mice at 3 days post infection (dpi) (Figure 1B), suggesting that Plscr1 is induced and potentially functions as an antiviral ISG in IAV infection *in vivo*.

Weight loss generally started at 3 dpi for both mouse strains. *Wt* infected mice reached their lowest weights around 8 dpi, while *Plscr1*^{-/-} infected mice experienced continued weight loss for an additional 2 days. *Plscr1*^{-/-} mice exhibited significantly greater weight loss compared to *Wt* mice with both sublethal and lethal dose infection at multiple timepoints (Figure 1C and 1D).

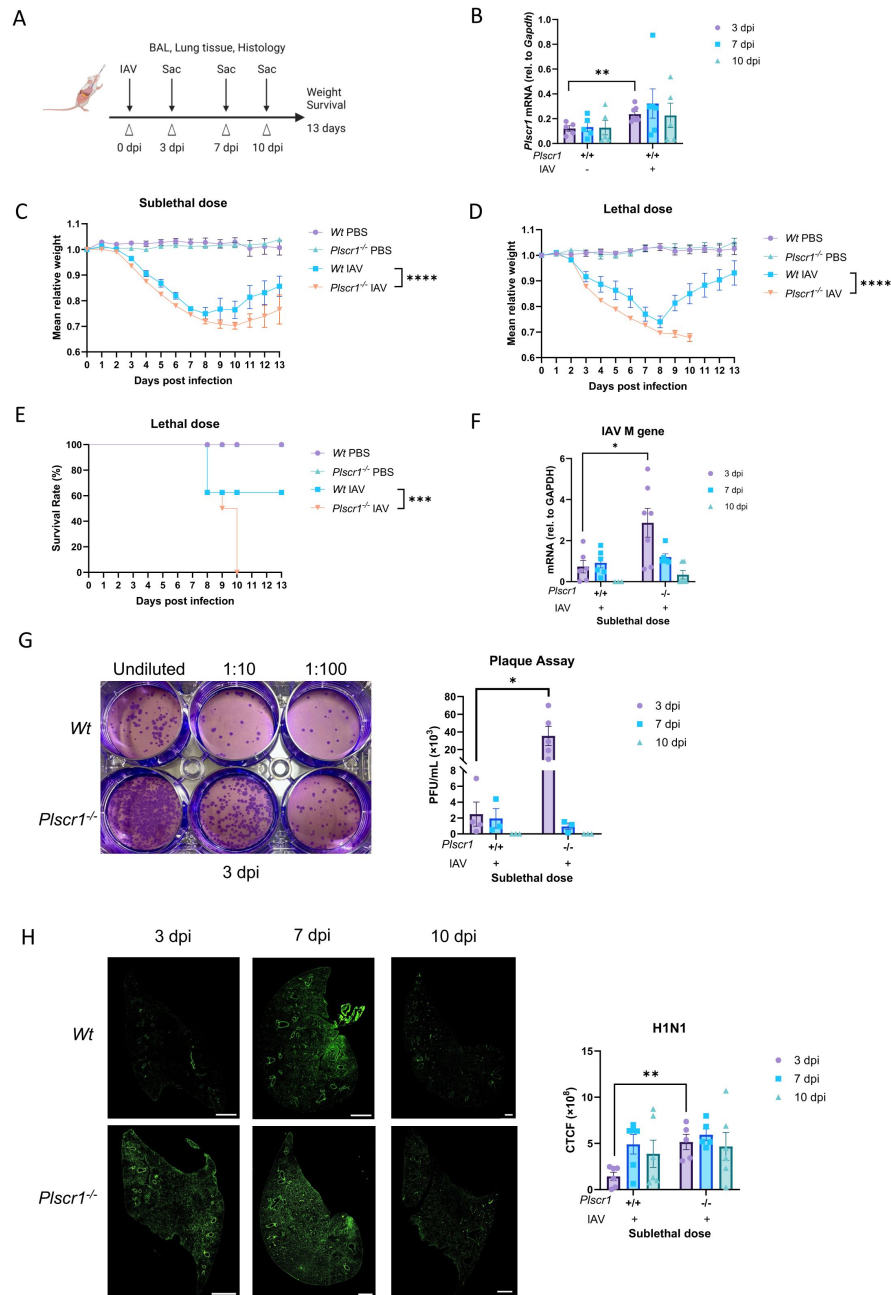


Figure 1.

Increased Susceptibility of *Plscr1*^{-/-} Mice to Influenza Virus Infection

Wt and *Plscr1*^{-/-} mice were exposed to sublethal (300 pfu, B, C and F-H) or lethal (900 pfu, D and E) IAV (WSN) infection. (A) Scheme of experiment. (B) Whole lungs of *Wt* mice were analyzed for *Plscr1* RNA by qRT-PCR. (C and D) Mean relative weight of mice post sublethal or lethal infection. (E) Survival rate of mice post lethal IAV infection. (F) Viral RNA load in the lungs was assessed by quantifying M gene by qRT-PCR. (G) Infectious viral titer in the lungs was assessed by plaque assays. (H) Representative staining for H1N1 in lungs. The scale bars represent 1 mm. Quantification was performed using ImageJ. Data are expressed as mean $\pm$ SEM of $n = 30$ mice/group for weight loss post sublethal infection and $n=8$ mice/group for weight loss and survival rate post lethal infection. For the rest analysis, $n = 5-10$ mice/group. All data were pooled from three independent experiments. Logrank (Mantel-Cox) test was used to compare survival rates. Ordinary two-way ANOVA tests were used to compare weight losses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. dpi, days post infection. CTCF, Corrected Total Cell Fluorescence.

Moreover, upon infection with 900pfu lethal dose of IAV, *Plscr1*^{-/-} mice had a significantly lower survival rate (**Figure 1E**). In fact, 100% of *Plscr1*^{-/-} mice died with 900 pfu of IAV in our experiment by 10 dpi.

We used three assays to robustly and comprehensively measure the viral burden in these animals. First, qRT-PCR analysis using cryopreserved lungs showed that IAV M gene segment mRNA was significantly higher in *Plscr1*^{-/-} lungs at 3 dpi (**Figure 1F**). This observation was further supported by the plaque assay, which measures infectious virus particles, and the IAV titer was significantly higher in *Plscr1*^{-/-} lungs at 3 dpi (**Figure 1G**). Finally, to visualize IAV burden and spread from upper respiratory tract, paraffin-embedded lung sections were stained for H1N1-FITC. In contrast to *Wt* lungs where IAV infection was mostly local and restricted to major airways, *Plscr1*^{-/-} lungs had viral dissemination extending from bronchioles to alveoli, with ~50% of lungs infected. Moreover, *Plscr1*^{-/-} lungs had significantly higher corrected total cell fluorescence (CTCF) than *Wt* lungs at 3 dpi, suggesting an overall higher viral burden during acute infection in the absence of *Plscr1* (**Figure 1H**).

Plscr1 Limits Innate Immunity-Mediated Inflammation and Lung Damage in IAV Infection

Bronchoalveolar lavage (BAL) was harvested for inflammatory cell counts using Cytospin followed by Diff-Quik stain. Immune cells infiltrated the lungs starting at 3 dpi and persisted at least until 10 dpi. *Plscr1*^{-/-} mice had significantly higher total BAL cell counts at 7 dpi compared to *Wt* mice, indicating a heightened inflammatory environment in alveoli during acute infection in the absence of *Plscr1* (**Figure 2A**). IAV attracted neutrophils in early infection (3-7 dpi) and lymphocytes in late infection (7-10 dpi). Interestingly, *Plscr1*^{-/-} lungs were significantly more neutrophilic at 3 dpi, and neutrophil populations persisted until 10 dpi (**Figure 2B**).

Lung sections were stained with Hematoxylin and Eosin (H&E) and histopathology was assessed. *Plscr1*^{-/-} lungs showed minor and localized tissue damage as early as at 3 dpi, when *Wt* lungs appeared normal. Consistently, aggravated immunopathology was evident in *Plscr1*^{-/-} lungs at 7 and 10 dpi compared to *Wt* lungs (**Figure 2C**). These observations included increased thickening and collapse of alveolar walls, pulmonary edema surrounding alveolar walls, inflammatory cell infiltration in peri-bronchial and parenchymal areas, and hyperemia in the absence of *Plscr1*, suggesting its role in maintaining tissue homeostasis in IAV infection.

To assess the role of *Plscr1* as an ISG, IFN expressions in whole lungs were measured by qRT-PCR. We found significantly elevated levels of *Ifn*- α , β , γ and λ expression in *Plscr1*^{-/-} mice, particularly in early infection (**Figure 2D**). This upregulation was associated with heightened production of *Tnf*- α and *Ifn*- λ in the BAL at 3 dpi, as assessed by ELISA (**Figure 2E**). These cytokines may implicate in hyperactive feedforward inflammatory circuits during early IAV infection leading to acute lung injury in *Plscr1*^{-/-} mice [24], consistent with worsened histopathology. On the contrary, the expressions of these mediators in *Wt* mice were well controlled, facilitating virus elimination without inciting excessive inflammation. These findings underscore the antiviral and potentially immunoregulatory role of *Plscr1*.

Plscr1 Binds to *Ifn*- λ 1 Promoter and Activates *Ifn*- λ 1 Transcription in IAV Infection

To determine if IFN signaling pathways are regulated by *Plscr1* during IAV infection, RNA sequencing of a total of 20,700 genes was performed using pooled samples from *Wt* and *Plscr1*^{-/-} mouse lungs infected with 300 pfu of IAV. A comprehensive examination of interferons and their receptors revealed that *Ifn*- λ 1 expression was significantly lower in *Plscr1*^{-/-} mice, despite high expression of *Ifn*- λ at both 3 and 7 dpi (**Figure 3A**). Importantly, *Ifn*- λ 1 was the only IFN receptor exhibiting this pattern, as the expression of its coreceptor *Il10-r* β and other IFN receptors

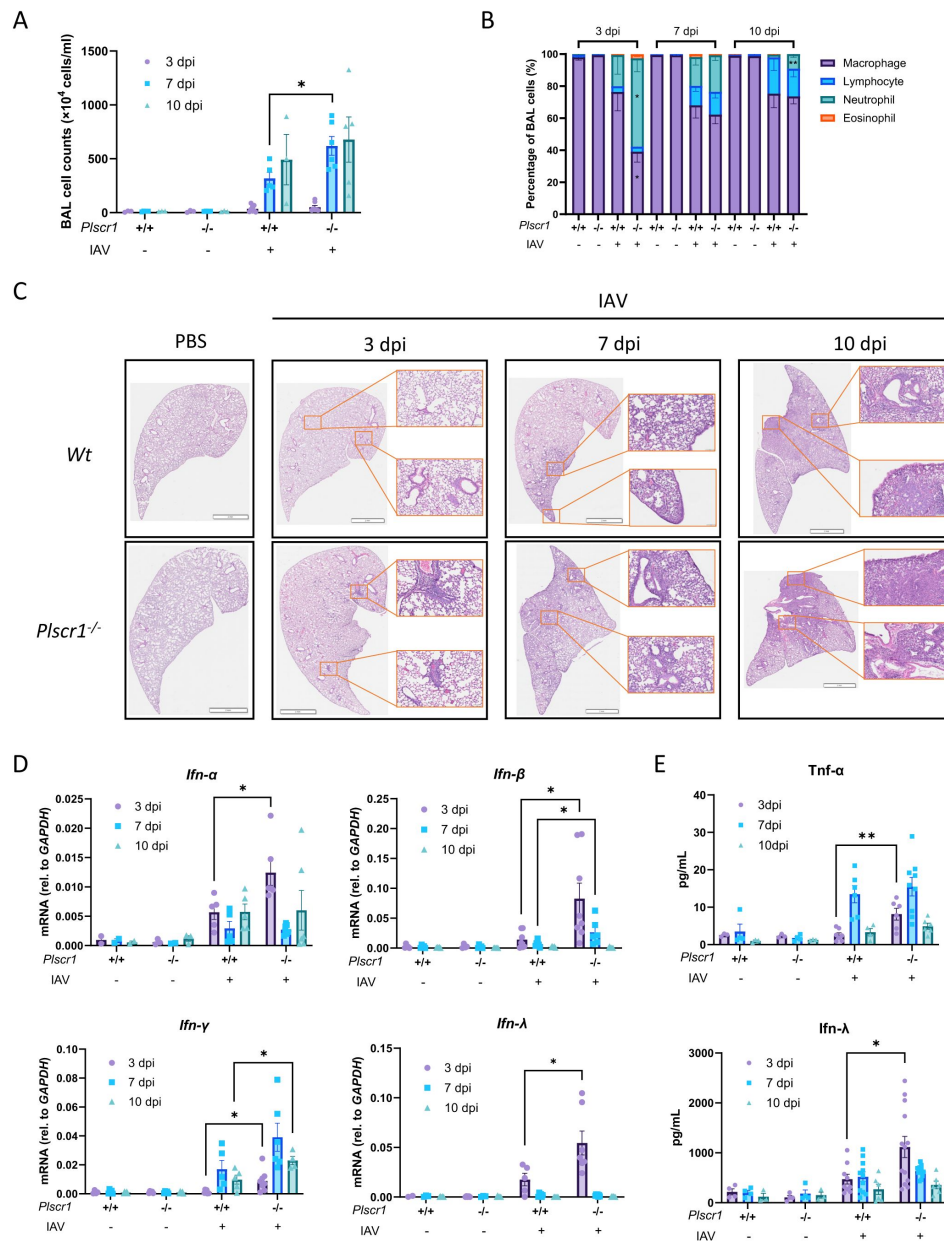


Figure 2.

Increased Lung Inflammation in *Plscr1*^{-/-} Mice in Influenza Virus Infection

Wt and *Plscr1*^{-/-} mice were exposed to sublethal (300 pfu) IAV (WSN) infection. (A) Total BAL leukocyte numbers. (B) Differential cell counts in BAL. (C) Representative lung sections stained with H&E. Scale bars represent 3 mm (main) and 200 μ m (inlays). (D) Whole lungs were analyzed for *Ifn-α*, *Ifn-β*, *Ifn-γ* and *Ifn-λ* RNA by qRT-PCR. (E) Tnf-α and *Ifn-λ* concentrations in BAL by ELISA. Data are expressed as mean $\pm$ SEM of $n = 3$ -14 mice/group. All data were pooled from three independent experiments. * $p < 0.05$, ** $p < 0.01$. dpi, days post infection.

remained unaltered. Inflammatory cytokines including type 1 and 2 IFNs, *Tnf-α* and *Il-1b* were also highly expressed in *Plscr1*^{-/-} infected mice, suggesting upstream cytokine production pathways were intact. The RNA sequencing result was further validated by qRT-PCR, which showed that *Plscr1*^{-/-} mice failed to upregulate *Ifn-λr1* expression at both 3 and 7 dpi compared to *Wt* mice (**Figure 3B** [↗](#)). Furthermore, using the online Interferome database[25 [↗](#)], we identified a total of 1,113 ISGs in our dataset with a fold change ≥2 (**Figure 3C** [↗](#)). Enlarged heatmaps with gene names are provided in Supplemental Figure 1. Among those ISGs, 584 are regulated exclusively by type 1 IFNs, and 488 are regulated by both type 1 and type 2 interferons. Unfortunately, the Interferome database does not include information on type 3 IFN-inducible genes in mice. Although many ISGs were robustly upregulated in *Plscr1*^{-/-} infected lungs, consistent with inflammation data, a large subset of ISGs failed to be transcribed when *Ifn-λr1* function was impaired, especially at 7 dpi. We suspect that those non-transcribed ISGs in *Plscr1*^{-/-} mice may be specifically regulated by type 3 IFN and represent interesting targets for future research. As a result, disrupted type 3 interferon signaling due to impaired expression of *Ifn-λr1* may underlie the delayed viral clearance and exacerbated immunopathology observed in *Plscr1*^{-/-} mice[10 [↗](#)].

To corroborate the cell-specific localization of *Ifn-λr1* expression, *Wt* IAV-infected mouse lungs were probed with immunofluorescent antibodies to detect *Ifn-λr1* and Foxj1 (a marker for ciliated epithelial cells), Uteroglobin (a marker for club cells) or Sftpc (a marker for type 2 alveolar epithelial (AT2) cells). Consistent with previous studies, *Ifn-λr1* was present on all of these epithelial cell types (**Figure 3D** [↗](#)). We further quantified *Ifn-λr1* expression in selected airways by measuring mean intensity to disregard the differences in airway areas. *Ifn-λr1* expression in alveoli were measured using Corrected Total Cell Fluorescence (CTCF), so that all fluorescent signals within the same size of alveolar areas were analyzed. Expression of *Ifn-λr1* was significantly reduced in *Plscr1*^{-/-} mouse lungs, in both airways and alveoli regions (**Figure 3E** [↗](#)). Calu-3 epithelial cells were then used to investigate if *PLSCR1* expression can be regulated by IFN-λ. Using qRT-PCR, we found *PLSCR1* transcription was significantly increased when stimulated with IFN-λ, and this effect was attenuated by pre-incubation with α-hIFNLR1 neutralizing antibody (**Figure 3F** [↗](#)). Additionally, IFN-λ could directly stimulate the expression of its receptor, *IFN-λR1*, thereby establishing a positive feedback loop to further drive the expression of *PLSCR1* (**Figure 3G** [↗](#)). These studies demonstrate that *PLSCR1* is an ISG that can be directly stimulated by IFN-λ in airway epithelial cells.

We subsequently investigated the mechanism underlying *Plscr1*'s transcriptional regulation of *Ifn-λr1* in airway epithelial cells. Chromatin-immunoprecipitation (ChIP) followed by standard PCR in IAV-infected Calu-3 cells unveiled that *Plscr1* physically bound to the promoter region of *Ifn-λr1*, with this binding becoming more evident in IAV-infected cells (**Figure 3H** [↗](#)). These results were further validated using real-time quantitative PCR, suggesting that *Plscr1* translocated into the nucleus of lung epithelial cells upon IAV infection to activate *Ifn-λr1* transcription (**Figure 3I** [↗](#)).

To further determine the regulation of *Ifn-λr1* by *Plscr1* in response to *Ifn-λ* signaling without the complexities associated with live virus infection, high molecular weight poly(I:C) was administered intranasally to mice daily for 6 days at 2.5mg/kg of body weight, and mice were sacrificed on the following day (Supplemental Figure 2A). We observed an increase in total BAL cell counts with poly(I:C) administration, although there was no difference between *Wt* and *Plscr1*^{-/-} mice (Supplemental Figure 2B). However, *Plscr1*^{-/-} mice exhibited a significantly higher number of lymphocytes in BAL (Supplemental Figure 2C), indicating that repetitive poly(I:C) administration might activate adaptive immune responses rather than the innate immune system in an acute IAV infection (**Figure 2B** [↗](#)). All interferon expressions were elevated following poly(I:C) administration, with comparable levels observed between *Wt* and *Plscr1*^{-/-} mice (Supplemental Figure 2D). Consistently, lung histopathology exhibited similar levels of inflammation or tissue damage in both *Wt* and *Plscr1*^{-/-} mice (Supplemental Figure 2E). Importantly, *Plscr1* expression was significantly induced by poly(I:C) (Supplemental Figure 2F).

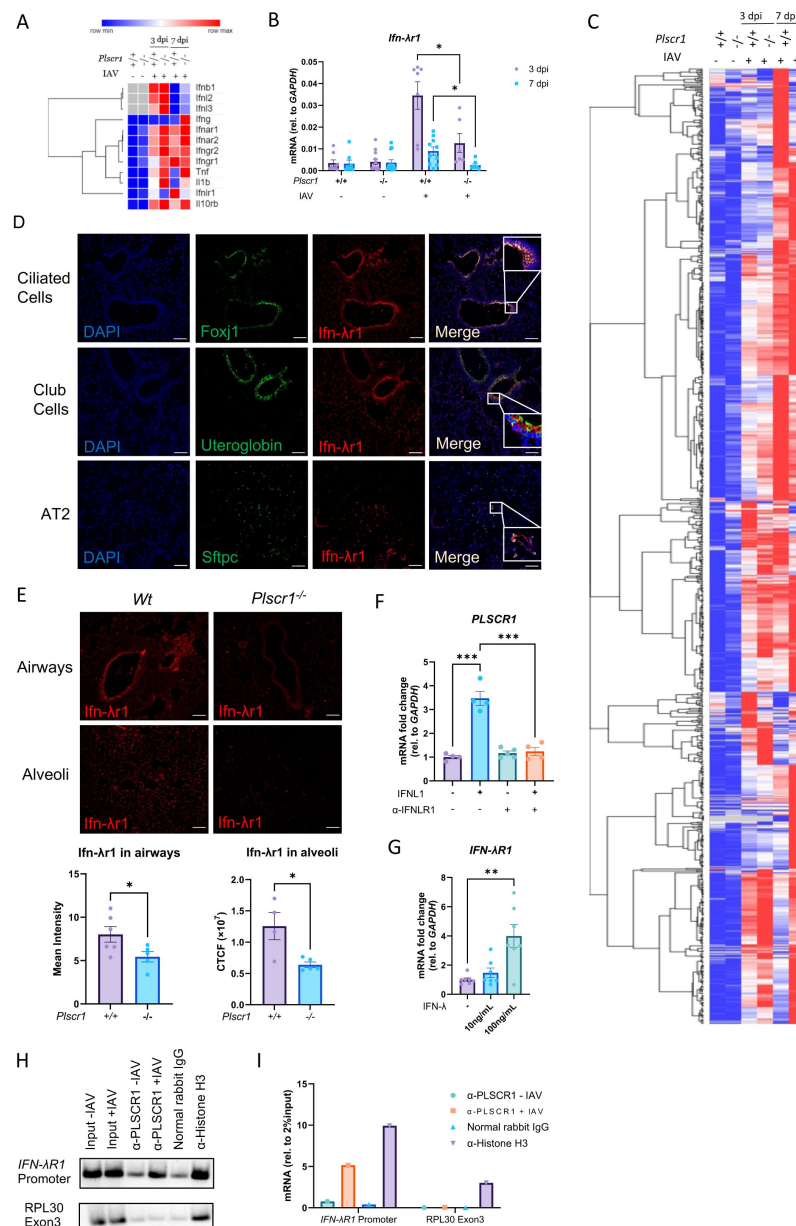


Figure 3.

Transcriptional Regulation of *IFN-λ1* by *PLSCR1* and IFN-λ in IAV Infection

(A-E) *Wt* and *Plscr1*^{-/-} mice were exposed to sublethal (300 pfu) IAV (WSN) infection. (A) Heatmap of interferons and their receptors in whole lungs by RNA-seq. (B) Whole lungs were analyzed for *Ifn-λr1* by qRT-PCR. (C) Heatmap of differential expressions of all ISGs in whole lungs by RNA-seq. Gene expressions were compared between groups within each row and color-labeled from row minimum (blue) to row maximum (red). (D) Localization of *Ifn-λr1*+ cells in the lungs of IAV-infected *Wt* mice at 7 dpi. Sections stained for *Ifn-λr1* (red), Foxj1, uteroglobin or Sftpc (green) and DAPI (blue) are shown. (E) Representative staining for *Ifn-λr1* in airways or alveoli of IAV-infected *Wt* and *Plscr1*^{-/-} mice at 7dpi. Quantifications were performed using ImageJ. (F-G) Calu-3 cells were analyzed for *PLSCR1* (F) and *IFN-λr1* (G) RNA by qRT-PCR after recombinant IFN-λ and/or α-IFN-λR1 antibody treatment. Data are presented as fold change compared to non-treated group. (H-I) Chromatin-Immunoprecipitation of *PLSCR1* and *IFN-λr1* promoter in Calu-3 cells followed by standard PCR (H) and real-time quantitative PCR (I). Data are expressed as mean ± SEM of n = 4-12 mice or wells/group. For transcriptomic analysis, 9 mice from each PBS-treated group and 4 mice from each IAV-infected group were pooled together. All data were pooled from three independent experiments. *p < 0.05, **p < 0.01, ***p < 0.001. dpi, days post infection. CTCF, Corrected Total Cell Fluorescence. Scale bars represent 50 μm.

Notably, although no other phenotypical or genotypical variances were observed in poly(I:C)-treated *Plscr1*^{-/-} mice, *Ifn-λr1* expression was significantly lower in these mice, further affirming the requirement of *Plscr1* for *Ifn-λr1* expression in response to IFN-λ (Supplemental Figure 2G).

Plscr1 Interacts with Ifn-λr1 on Pulmonary Epithelial Cell Membrane in IAV Infection

We employed a combination of multiple biophysical approaches to fully assess the potential protein interaction between *Plscr1* and *Ifn-λr1*. Co-immunoprecipitation (Co-IP) followed by western blot demonstrated that *Plscr1* successfully pulled down *Ifn-λr1* in both uninfected and IAV-infected *Wt* mouse lungs (Figure 4A and Supplemental Figure 3). Since the co-IP assay may detect indirect interactions through intermediate partners, we used proximity ligation assay (PLA) as a more precise method to identify direct interactions. Consistently, using PLA (Figure 4B), we detected direct interactions between *Ifn-λr1* and *Plscr1* on airway and alveolar epithelial cells in *Wt* mouse lungs. Noteworthy, significantly more and stronger PLA signals per area were detected in IAV-infected lungs compared to uninfected lungs. In agreement with the PLA, in unpermeabilized Calu-3 cells, *Plscr1* and *Ifn-λr1* colocalized on the cell membrane after IAV infection (Figure 4C). In contrast, neither colocalization nor clear expression of *Plscr1* or *Ifn-λr1* was evident without infection, implying an infection-specific interaction in human airway epithelial cells (Figure 4C). Hence, *Plscr1* may regulate *Ifn-λr1* expression and IFN-λ signaling in airway epithelial cells through both gene transcription and protein interaction in IAV infection.

Both Cell Surface and Nuclear PLSCR1 Regulates IFN-λ Signaling and Limits IAV Infection Independent of Its Enzymatic Activity

PLSCR1 mutants with specific PLSCR1 cellular distribution were employed to determine the relative contributions of cell surface and nuclear PLSCR1 in regulating IFN-λ signaling and IAV infection. PLSCR1 contains a 5-cysteine palmitoylation motif (C¹⁸⁴CCPCC¹⁸⁹), where substitution of these cysteines with alanine (labeled 5CA) completely abolishes its membrane localization, leading to exclusive localization in the cytosol and nucleus[26]. On the other hand, a single amino acid mutation of histidine²⁶² to tyrosine (labeled H262Y) in the non-classical nuclear localization signal of PLSCR1 completely abates its nuclear localization, leaving PLSCR1 exclusively in the cytosol and on the cell membrane[27]. In addition, given that PS externalization plays a role in cell death regulation[19], we asked whether the enzymatic activity of PLSCR1 interferes with IFN-λ signaling. Phenylalanine²⁸¹ in the β-barrel hydrophobic loop region of PLSCR1 has been shown to be important for Ca²⁺-dependent PS exposure[27], and a substitution with alanine (labeled F281A) renders PLSCR1 enzymatically inactive [15].

PLSCR1(WT), *PLSCR1(5CA)*, *PLSCR1(H262Y)* and *PLSCR1(F281A)* plasmids built on PLV-EF1a-IRES-Hygro backbone were amplified in DH5α competent cells, co-transfected with HIV lentivector into 293T cells, and transduced into *Plscr1*^{-/-} A549 cells. After a 10-day hygromycin selection, transduction efficiency and subcellular locations of PLSCR1 were analyzed using flow cytometry (Supplemental Figure 4A). *PLSCR1(WT)* exhibited higher PLSCR1 expression than *Plscr1*^{+/-} cells, indicating overexpression resulting from transduction (Supplemental Figure 4B). Consistent with previous findings, *PLSCR1(5CA)* exhibited low surface expression, and *PLSCR1(H262Y)* had low nuclear expression of PLSCR1. *PLSCR1(F281A)* showed a similar distribution compared to *PLSCR1(WT)*, confirming that the loss of enzymatic activity did not restrict the subcellular localization. Transduction efficiency ranged between 37% to 70% (Supplemental Figure 4C).

In *PLSCR1*^{+/-} cells or cells transduced with *PLSCR1(WT)*, *PLSCR1(5CA)* or *PLSCR1(F281A)*, IFN-λR1 expression was significantly induced by IAV infection. We found that PLSCR1 nuclear import was required for IFN-λR1 expression, as *IFN-λR1* transcription and protein translation was not induced in cells transduced with *PLSCR1(H262Y)* or in *PLSCR1*^{-/-} cells (Figure 5A and 5B). In contrast,

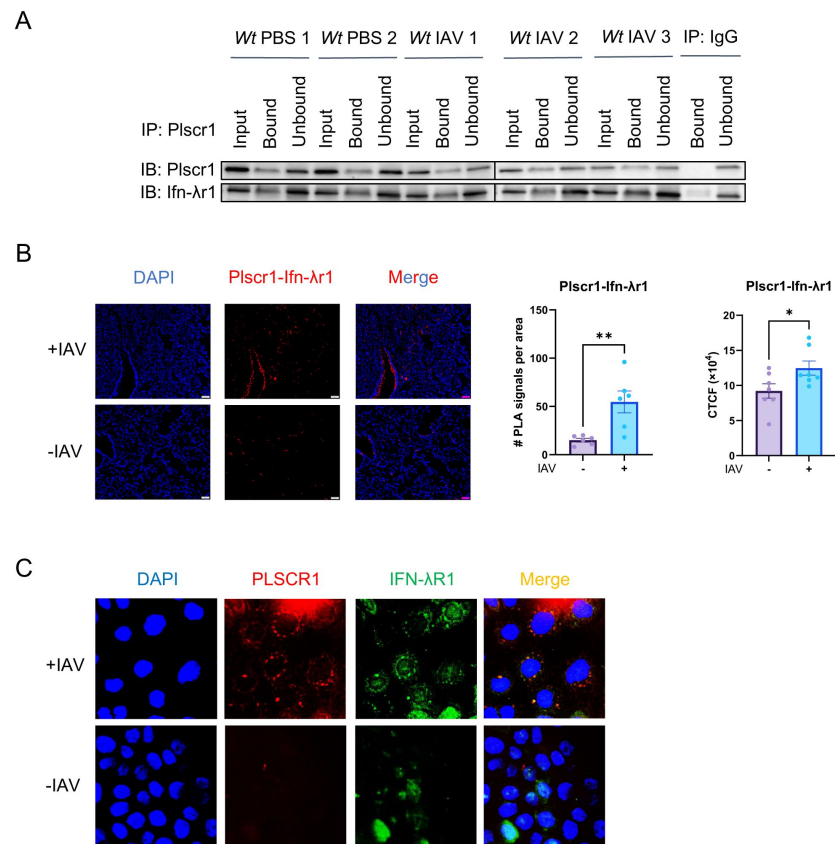


Figure 4.

Protein Interaction between IFN-λR1 and PLSCR1 in IAV Infection

(A) Co-Immunoprecipitation of Plscr1 and Ifn-λr1 in whole mouse lungs followed by western blot. (B) Proximity ligation assay of Ifn-λr1 and Plscr1 in the lungs of *Wt* mice infected or uninfected with IAV. Scale bars represent 50 μm. Quantifications were performed using ImageJ. (C) Colocalization of IFN-λR1 (green) and PLSCR1 (red) on Calu-3 cell membranes infected or uninfected with IAV in a nonpermeabilized staining. Scale bars represent 10 μm. Data are expressed as mean ± SEM of n = 6-7 lungs/group. All data were pooled from three independent experiments. *p < 0.05, **p < 0.01. PLA, Proximity Ligation Assay. CTCF, Corrected Total Cell Fluorescence.

minimal protein interactions between PLSCR1 and IFN- λ R1 were detected by PLA in cells transduced with *PLSCR1(5CA)* or in *PLSCR1*^{-/-} cells, while *PLSCR1*^{+/+} cells and cells transduced with *PLSCR1(WT)*, *PLSCR1(H262Y)* and *PLSCR1(F281A)* had strong protein binding of PLSCR1-IFN- λ R1, demonstrating that PLSCR1 interacts with IFN- λ R1 on cell membrane (**Figure 5C**). Furthermore, the lipid scramblase activity of PLSCR1 is uncoupled from its regulation of IFN- λ signaling.

When compared to *PLSCR1*^{+/+} cells, *PLSCR1(WT)*, *PLSCR1(5CA)* and *PLSCR1(F281A)* provided similar protection against IAV infection. However, *PLSCR1(H262Y)* showed only partial viral clearance, as demonstrated by viral copy numbers measured by M gene segment qRT-PCR quantification (**Figure 5D**) and infectious viral titers from plaque assays (**Figure 5E**). This is further validated with a cell coverage assay when a high dose IAV was used for infection. Since the proliferation between differently transduced cell lines was similar and all cells were grown to full confluency before infection, the cell coverage represented resistance to infection and survival of cells. While *PLSCR1(5CA)* and *PLSCR1(F281A)* had a high surface coverage comparable to *PLSCR1(WT)*, *PLSCR1(H262Y)* lost about half of the coverage (**Figure 5F**). Therefore, nuclear PLSCR1 is both essential and sufficient for IAV control through *IFN- λ R1* transcription regulation. Nevertheless, *PLSCR1(H262Y)* still rescued a significant number of cells when reintroduced into *PLSCR1*^{-/-} cells, indicating a partial protective role of PLSCR1 on epithelial cell surface through IFN- λ R1 protein regulation. Moreover, the anti-IAV function of PLSCR1 is independent of its lipid scramblase activity. Importantly, our findings that the scramblase activity is not required for protection against IAV infection is consistent with earlier observations with SARS-CoV-2 [15], suggesting that the scramblase activity of PLSCR1 may generally be dispensable for protection against major human viruses.

The Anti-Influenza Activity of Plscr1 Is Highly Dependent on Ifn- λ r1

To establish a causal link between the impaired type 3 IFN pathway and the increased susceptibility to IAV observed in *Plscr1*^{-/-} mice and *PLSCR1*^{-/-} cells, *Ifn- λ r1*^{-/-} mice were crossed with *Plscr1*^{-/-} mice to generate double-knockout *Plscr1*^{-/-}*Ifn- λ r1*^{-/-} mice. Immunofluorescence confirmed the absence of Ifn- λ r1 expression in *Ifn- λ r1*^{-/-} mice (Supplemental Figure 5A). Following sublethal IAV infection, *Ifn- λ r1*^{-/-} mice exhibited significantly greater body weight loss at 3 dpi than *Plscr1*^{-/-} mice, which retain basal Ifn- λ r1 expression (Supplemental Figure 5B). Moreover, they also showed significantly higher total BAL cell counts (Supplemental Figure 5C), increased neutrophil percentages (Supplemental Figure 5D), and higher infectious viral titers as measured by plaque assays (Supplemental Figure 5E). These findings indicate that complete loss of Ifn- λ r1 results in greater susceptibility to IAV than the loss of *Plscr1*-mediated Ifn- λ r1 upregulation, underscoring the essential role of the type 3 IFN pathway in anti-viral defense.

Importantly, *Plscr1*^{-/-}*Ifn- λ r1*^{-/-} mice showed no further increase in weight loss (Supplemental Figure 5B), total BAL cell counts (Supplemental Figure 5C), neutrophil percentages (Supplemental Figure 5D), and IAV titers (Supplemental Figure 5E) compared to *Ifn- λ r1*^{-/-} mouse lungs, indicating that the antiviral activity of *Plscr1* is largely dependent on Ifn- λ r1.

PLSCR1 Expression Is Upregulated in the Ciliated Airway Epithelial Compartment of Mice following Flu Infection

To understand the specific cell types that have increased PLSCR1 expression following flu infection *in vivo*, we explored a comprehensive scRNA sequencing dataset generated from uninfected or IAV-infected mouse lungs at 0, 1, 3, 6 and 21 dpi [28]. Distinct lung cell populations were identified using the Seurat algorithm. We assigned a total of 38 clusters based on their transcriptomic signatures, including 13 epithelial populations, 5 endothelial populations, 9 mesenchymal populations and 11 immune populations (Table S2). Two-dimensional UMAP (**Figure 6A**) and bar charts of the proportion (Supplemental Figure 6A) and cell count (Supplemental Figure 6B) of each cluster demonstrated that these main lung cell populations (epithelial, endothelial, mesenchymal, and immune) were dynamic over the course of infection. Following

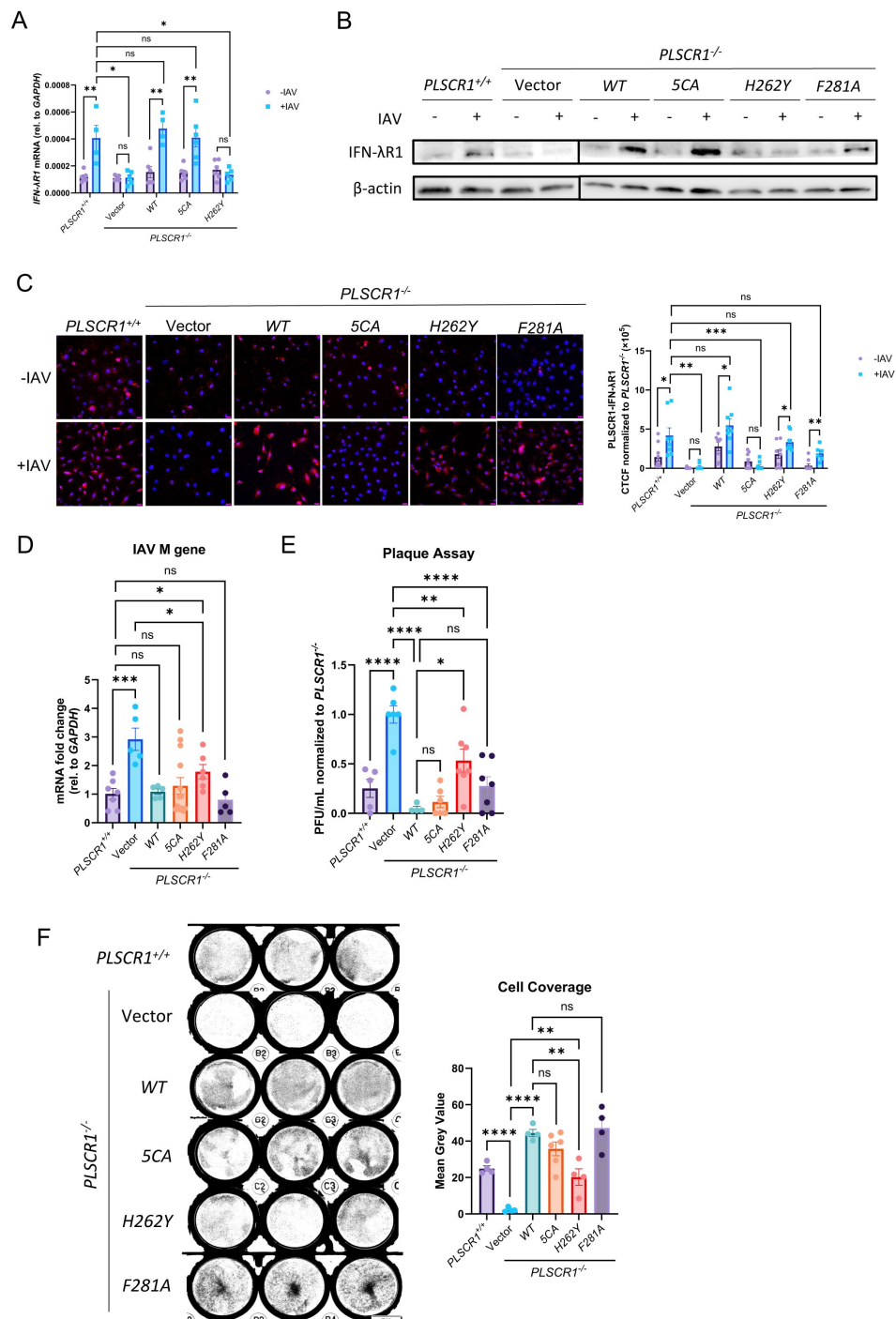


Figure 5.

Requirement of Both Nuclear and Surface PLSCR1 but Not the Enzymatic Activity in IFN-λR1-Mediated Anti-Influenza Activities

PLSCR1^{-/-} A549 cells were transduced with mutated *PLSCR1* plasmids using lentivirus and infected with IAV (PR8) for 24 hours at 1 MOI (A-E) or 10 MOI (F). (A) *IFN-λR1* RNA by qRT-PCR. (B) IFN-λR1 proteins by western blotting. (C) Proximity ligation assay of IFN-λR1 and PLSCR1. Scale bars represent 20 μm. Quantifications were performed using ImageJ. (D) Viral RNA load was assessed by quantifying M gene by qRT-PCR. (E) Infectious viral titer was assessed by plaque assays. (F) Cells were stained with crystal violet. Cell viability was quantified using ImageJ. Data are expressed as mean ± SEM of n = 4-13 wells/group. All data were pooled from three independent experiments. ns, not significant, *p < 0.05, **p < 0.01, ***p < 0.001, ****p < 0.0001, *****p < 0.00001. CTCTF, Corrected Total Cell Fluorescence.

infection, many populations emerged, particularly within the immune cell clusters. At the same time, some clusters were initially depleted and later restored, such as microvascular endothelial cells (cluster 2). Other populations, such as interferon-responsive fibroblasts (cluster 20), showed a dramatic yet transient expansion during acute infection and disappeared after infection resolved.

Within the epithelial populations, *Plscr1* was mainly expressed by ciliated epithelial cells, mesothelial epithelial cells, Epcam+Pecam1+ cells, club cells and AT1 cells (in decreasing order of aggregated expression, **Figure 6B**). Conversely, AT2 cells, epithelial-mesenchymal transitional cells and Krt8+ cells exhibited very low levels of *Plscr1* expression. Among the various epithelial populations, ciliated epithelial cells exhibited both the highest aggregated expression of *Plscr1* (**Figure 6B**) and the most significant upregulation ($p < 2.22 \times 10^{-16}$ and $p = 6.7 \times 10^{-5}$) at 3 dpi in early IAV infection (**Figure 6C** and Supplemental Figure 7A-7K). In contrast, AT1 cells were the only other epithelial cluster to show *Plscr1* upregulation at 3dpi, but to a much less extent ($p = 0.033$, Supplemental Figure 7J). Moreover, the increased expression of *Plscr1* persisted for at least 21 days in ciliated epithelial cells, suggesting that *Plscr1* may primarily exert its anti-flu activities in these cells (**Figure 6C**).

For the predominant ciliated epithelial cell cluster, we performed Gene Ontology (GO) analysis of the differentially expressed genes to compare potentially distinct functions before and after infection (**Figure 6D**). We found that the most upregulated pathways at 1 dpi were related to protein folding and localization, indicating a rapid cellular response to infection by altering their protein profile. At 21 dpi, ciliated epithelial cells were characterized by adaptive immune regulation, such as antigen presentation and memory cell generation. At 3 and 6 dpi, ciliated epithelial cells exhibited similar innate immune and inflammatory signatures dominated by interferon signaling. As expected, *Plscr1* participated in all of the top 10 enriched pathways, implying that it could facilitate robust antiviral responses in ciliated epithelial cells as an ISG during acute infection.

Gene expression levels of the most differentially expressed genes at individual timepoints in ciliated epithelial cells were shown in a heatmap (**Figure 6E**). Signature genes for cells at homeostasis included *Jund* (JunD proto-oncogene), *mt-Nd3* (mitochondrial NADH dehydrogenase 3) and *Cdkn1c* (cyclin-dependent kinase inhibitor 1C), which are regulators of cell metabolism and proliferation[29–31]. Signature genes enriched at 1 dpi were *Hspa5* (heat shock protein family A member 5), *Hsph1* (heat shock protein family H member 1) and *Clu* (clusterin), which play critical roles in protein folding and assembly[32–34]. Consistent with the GO analysis, ciliated cells at 3 and 6 dpi had very similar signature genes associated with interferon pathways, namely a large component of ISGs, such as *Isg15* (interferon-stimulated gene 15), *Oasl2* (2'-5'-oligoadenylate synthetase like 2) and *Irf7* (interferon regulatory factor 7). Importantly, *Plscr1* is required to potentiate the expressions of many of these ISGs[23], highlighting its potentially multifunction in anti-flu responses of ciliated epithelial cells. Signature genes for 21 dpi largely overlapped with those for 10 dpi, containing many histocompatibility 2, class II antigen proteins, which are mediators of MHC class II antigen processing and presentation.

In contrast to epithelial clusters, immune populations had low levels of *PLSCR1*, with relatively high expressions observed in alveolar macrophages, NK cells and regulatory T cells (**Figure 6F**). However, we did not observe any significant difference in *PLSCR1* levels before and post-infection in macrophages and neutrophils (**Figure 6G** and **6H**), the two immune populations that express IFN- λ R1 besides dendritic cells[35]. Hence, the anti-influenza activities we described are unlikely to be carried out by immune cells.

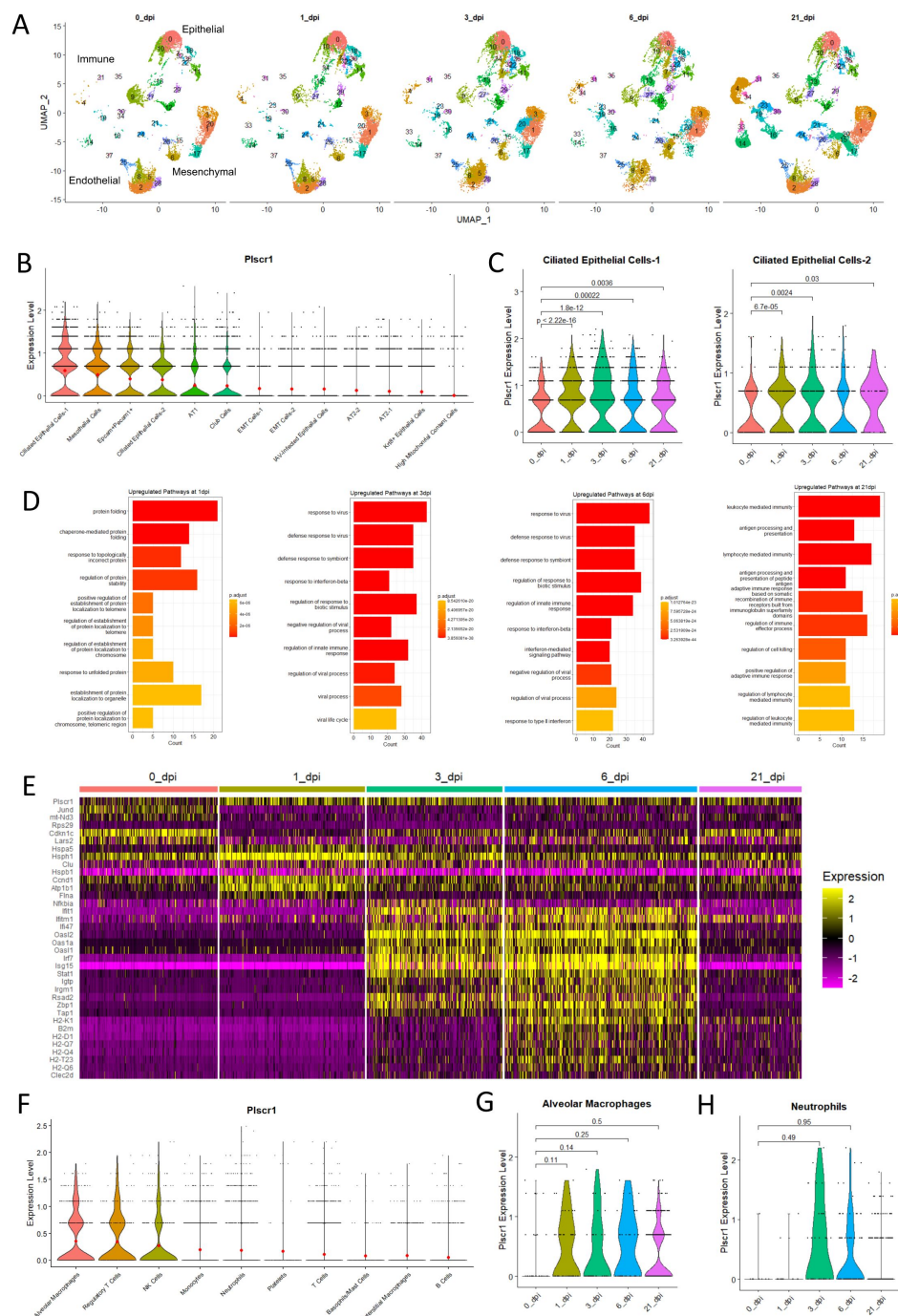


Figure 6.

Cell-Specific Roles of *Plscr1* in Influenza Virus Infection in Mice

Wt mice were exposed to 2500 EID₅₀ IAV (PR8) infection. Lungs were used for single-cell RNA sequencing analysis at 0, 1, 3, 6 and 21 dpi. (A) Two-dimensional UMAP representation of individual cells obtained from different timepoints. (B) Violin plot of aggregated *Plscr1* expressions in all epithelial cell clusters. Red dots represent mean expression levels. (C) Violin plot of time-dependent *Plscr1* expressions in both ciliated epithelial cell clusters. (D) GO analysis for upregulated pathways in Ciliated Epithelial Cells-1. (E) Heatmap of the most differentially expressed genes of Ciliated Epithelial Cells-1 at different timepoints. (F) Violin plot of aggregated *Plscr1* expressions in all immune cell clusters. Red dots represent mean expression levels. (G) Violin plot of time-dependent *Plscr1* expressions in alveolar macrophage cluster. (H) Violin plot of time-dependent *Plscr1* expressions in neutrophil cluster.

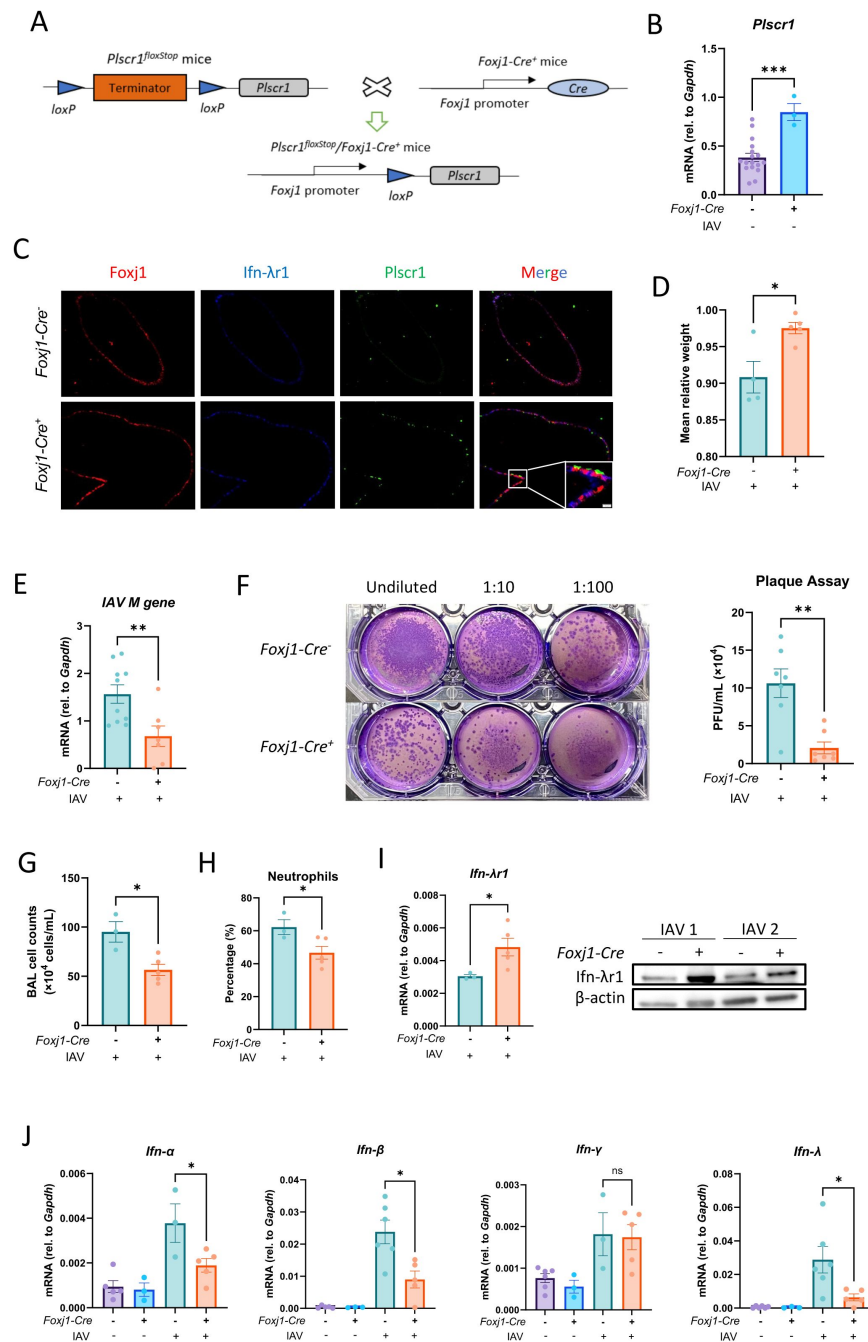


Figure 7.

Reduced Susceptibility of *Plscr1^{floxStop}/Foxj1-Cre⁺* Mice to Influenza Virus Infection

Plscr1^{floxStop} and *Plscr1^{floxStop}/Foxj1-Cre⁺* mice were exposed to sublethal (300 pfu) IAV (WSN) infection and sacrificed at 3 dpi. (A) Schematic representation of the experimental design of ciliated epithelial cell conditional *Plscr1* *KI* mice. (B) Validation of *Plscr1* overexpression in lungs of *Plscr1^{floxStop}/Foxj1-Cre⁺* mice by qRT-PCR. (C) Representative immunofluorescent staining for *Plscr1*, *Ifn-λr1* and *Foxj1* in lungs. (D) Mean relative weight of mice. (E) Viral RNA load in the lungs was assessed by quantifying M gene by qRT-PCR. (F) Infectious viral titer in the lungs was assessed by plaque assays. (G) Total BAL leukocyte numbers. (H) Neutrophil percentages in BAL. (I) Whole lungs were analyzed for *Ifn-λr1* RNA by qRT-PCR and *Ifn-λr1* protein by western blot. (J) Whole lungs were analyzed for *Ifn-α*, *Ifn-β*, *Ifn-γ*, *Ifn-λ* RNA by qRT-PCR. (K) Model depicting proposed mechanism of PLSCR1-regulated IFN-λ signaling. Data are expressed as mean ± SEM of n = 3-10 mice/group. ns, not significant, *p < 0.05, **p < 0.01, ***p < 0.001. dpi, days post infection.

Overexpression of *Plscr1* in Ciliated Epithelial Cell, but Not Myeloid Cell, Is Sufficient to Provide Anti-Flu Protection through IFN- λ Signaling

To further validate the relevant contributions of ciliated epithelial cells versus myeloid populations including macrophages and neutrophils, we generated *Rosa26* locus targeted *Plscr1* conditional knock-in transgenic mice (*Rosa26-LoxP-STOP-LoxP-Plscr1* Tg; labeled *Plscr1^{floxStop}*). These mice were bred with *Foxj1-Cre* or *LysM-Cre* mice to overexpress *Plscr1* in ciliated epithelial cells or myeloid cells respectively (Figure 7A). Overexpression was confirmed by qRT-PCR with whole lung lysate and immunofluorescence staining at baseline (Figure 7B, 7C and Supplemental Figure 8A). At 3 dpi with sublethal IAV infection, *Plscr1^{floxStop}Foxj1-Cre⁺* mice lost less body weight (Figure 7D). Moreover, they had lower viral copy numbers as measured by IAV M gene segment qRT-PCR (Figure 7E), and lower infectious viral titer measured by plaque assays (Figure 7F) compared to *Plscr1^{floxStop}* mice, indicating overexpression of *Plscr1* in ciliated epithelial cells provides protection against flu infection. Concurrently, they had significantly lower total BAL cell counts (Figure 7G), lower neutrophil percentages (Figure 7H) and lower type 1 and 3 *Ifn* expressions (Figure 7J). We then examined *Ifn- λ 1*, and found that its transcription and protein expression were further increased in *Plscr1^{floxStop}Foxj1-Cre⁺* mice during acute infection (Figure 7I). Therefore, ciliated epithelial cell-specific overexpression of *Plscr1* is sufficient to provide anti-influenza protection through IFN- λ signaling. In contrast, using similar approaches, we found that overexpression of *Plscr1* in myeloid cells did not protect mice from IAV infection: *Plscr1^{floxStop}LysM-Cre⁺* mice exhibited comparable weight loss, total BAL cell counts, viral copy numbers, histopathology and expressions of interferons, *Ifn- λ 1* and *Plscr1* at both early and late timepoints examined (Supplemental Figure 8B-8I).

Taken together, consistent with the scRNA sequencing, transgenic mouse models with cell-specific overexpression of *Plscr1* demonstrated that *Plscr1* offers antiviral protection in ciliated airway epithelial cells through IFN- λ signaling (Figure 8).

Discussion

We are the first group to demonstrate the roles of *Plscr1* in a mouse-adapted human IAV-infected mouse model, to implicate its IFN- λ signaling-related mechanisms, and to elucidate the cell types that are responsible for *Plscr1*-mediated anti-influenza activities. We established *Plscr1^{-/-}* mice and found them more susceptible to IAV (WSN) compared to *WT* mice, as evidenced by greater weight loss in both sublethal and lethal infection and poorer survival in a lethal infection. Further examination of infected lungs provided the first *in vivo* evidence demonstrating that *Plscr1* suppressed human IAV replication. This observation aligns with previous report indicating that PLSCR1 interacts with the IAV NP, thereby impairing its nuclear import *in vitro*[20]. Notably, while differences in viral copy numbers were only observed at the early stages of infection, coinciding with a significant increase in *Plscr1* transcription, these changes had profound implications for host fitness. Therefore, as one of the earliest induced ISGs, *Plscr1* constitutes the frontline defense against influenza infection.

While the only previously published *Plscr1^{-/-}* mouse flu model focused on an H1N1 SIV infection[21], our data showed both similarities and discrepancies. First, while both studies observed that *Plscr1* promoted survival during IAV infection, SIV-infected *Plscr1^{-/-}* mice exhibited weight loss similar to *Wt* mice. Furthermore, while both models attributed the lower survival rate in *Plscr1^{-/-}* mice to increased viral replication, SIV-infected *Plscr1^{-/-}* lungs exhibited higher viral titers across all examined time points, from 1 to 7 dpi. Intriguingly, contrary to our observations, *Plscr1* expression was markedly decreased in SIV infection. Given previous *in vitro* studies

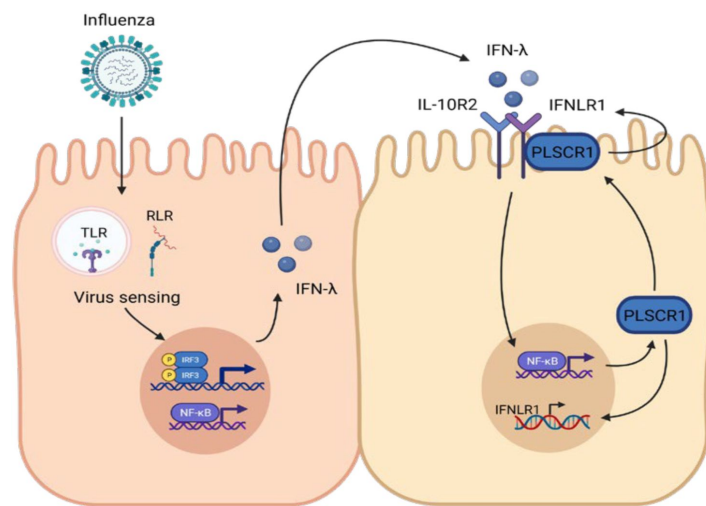


Figure 8.

Model depicting proposed mechanism of PLSCR1-regulated IFN-λ signaling.

Influenza infection is usually detected by intracellular pattern recognition receptors (PRRs) such TLR 3 and 7, RIG-I and MDA5. These PRRs activate the expression of *IFN-λ* in early infection stage through IRF-3 and NF-κB-controlled transcriptions. IFN-λ secreted by the infected cells interacts with IL-10R2 and IFN-λR1 on neighboring cell surface, which results in activation of expression of various IFN-stimulated genes, including *PLSCR1*. In ciliated airway epithelial cells, *PLSCR1* can further enhance the transcription of *IFN-λR1* by directly binding to its promoter region as a transcriptional factor, or interact with IFN-λR1 on the cell membrane.

demonstrating PLSCR1 induction by IAV (WSN)[20] and type 1 IFNs[13, 23, 36], we propose that the contradictory trend observed by Liu et al. may be attributed to distinct properties of SIV, such as viral replication rate, both the cellular tropism and the tissue tropism (proximal or distal lung), or antigen variation which may affect direct interaction with PLSCR1, innate sensing of the infection, or recognition by the adaptive immune response.

The delicate balance between immunity and immunopathology plays a pivotal role in determining host fitness during viral infections. To interrogate immunopathology in the lungs, we accessed the BAL, histology and interferon expressions. BAL from *Plscr1*^{-/-} mice were highly enriched with inflammatory neutrophils and lymphocytes, which were likely attracted by robust IFNs and other chemokines. Consistently, *Plscr1*^{-/-} mice exhibited more severe lung damage and a greater extent of affected areas. These findings indicate that *Plscr1* not only enhances immunity but also mitigates immunopathology. Importantly, regardless of excessive production of antiviral IFNs in *Plscr1*^{-/-} mice, they failed to effectively control the initial viral infection. This suggests that the absence of *Plscr1* impairs the IFN signaling pathway, highlighting the crucial role of *Plscr1* in facilitating effective antiviral responses.

Although type 1 and 3 IFNs may share similar downstream pathways, they rely on distinct receptors for signaling. Consistent with previous findings[3], *Ifn-λr1* was detected in respiratory epithelium, including ciliated epithelial cells, club cells and AT2 cells during infection. Loss of *Plscr1* impaired *Ifn-λr1* transcription in IAV infection, with this transcriptional difference translating into protein expression. IFN-λ is crucial for early viral control within the initial days of infection without igniting unnecessary inflammation and compromising host fitness[10]. With limited *Ifn-λr1* expression, *Plscr1*^{-/-} mice were unable to mount a robust type 3 IFN response to control early viral infection. Instead, they relied largely on type 1 interferons, which succeeded in eliminating IAV at later time points, but led to exaggerated immunopathology. Furthermore, our observations of enhanced neutrophilia, lung injury and lethality in *Plscr1*^{-/-} mice align with findings reported in *Ifn-λr1*^{-/-} mice in IAV infection [10]. However, a discrepancy in *Ifn-λr1* expression over the course of infection was observed between the RNA sequencing and the qRT-PCR data. While RNA-seq showed further upregulation of *Ifn-λr1* at 7 dpi (Figure 3A), qRT-PCR indicated a rapid downregulation at the same time point (Figure 3B). The reason for this time-dependent discrepancy remains unclear and warrants further investigation. In addition, this study did not definitively establish causality between reduced *Ifn-λ* signaling and the observed *in vivo* phenotype. The increased morbidity and mortality in *Plscr1*^{-/-} mice could also be attributed to elevated *Tnf-α* levels and associated lung damage. Given that proinflammatory cytokines and/or enhanced lung damage are known contributors to influenza morbidity and mortality, future work will be needed to disentangle the impacts of *TNF-α*, *IL-1β* and other inflammatory cytokines from those of the IFN pathway to fully clarify the role of *Plscr1* in antiviral defense.

PLSCR1 expression was increased in response to IFN-λ in human airway epithelial cells, consistent with previous studies[15]. While PLSCR1 typically localizes on the cell membrane and in the cytoplasm, it translocates into nucleus to bind the *IFN-λR1* promoter upon IAV infection, thereby regulating *IFN-λR1* transcription. The nuclear localization and functions of PLSCR1 have been extensively documented in previous studies[17, 37–39]. Relevantly, IFN-α promotes the nuclear translocation of PLSCR1 in breast cancer cells[26]. Therefore, it is highly plausible that the nuclear trafficking of PLSCR1 in airway epithelial cells is similarly stimulated by IFN-α produced during IAV infection, but further evidence is demanded. Additionally, the precise binding site for PLSCR1 within the *IFN-λR1* promoter and the binding motif on PLSCR1 remain unknown. Previous bioinformatics predictions revealed that the -430~421 segment of *IFN-λR1* promoter likely contains binding sites for a number of transcription factors (TFs) with important regulatory functions[40]. Further mutagenesis studies, such as truncations or single nucleotide mutations within these sequences, could be done to identify the specific motif for PLSCR1 binding. Finally, it is not clear whether PLSCR1 directly activates *IFN-λR1* expression by acting as a TF, or as

a co-factor enhancing other TF's transcriptional activity. Coimmunoprecipitations could be pursued in future to explore the binding potential between PLSCR1 and other known TFs for *IFN-λR1*, such as NF-Υ[40].

In addition to activities in the nucleus, PLSCR1 has been shown to interact with multiple proteins on the plasma or endosomal membrane[22, 41–44]. Here we reported a novel interaction between PLSCR1 and *IFN-λR1* on airway epithelial cell membrane *in vivo* and *in vitro*, confirmed with both coimmunoprecipitation and immunofluorescence. Since their interaction was significantly enhanced after IAV infection, we speculate that membrane-bound PLSCR1 is a positive regulator of *IFN-λR1* signaling. One plausible mechanism is that PLSCR1 facilitates the intracellular trafficking of *IFN-λR1*, akin to its role in assisting the trafficking of other membrane receptors[22, 42, 45].

The subcellular location of PLSCR1 is vital not only for interactions with host components, but also for direct viral control. We found that nuclear PLSCR1 is both necessary and sufficient for viral control in airway epithelial cells, whereas membrane PLSCR1 provides only partial protection against IAV infection. These findings are not surprising, as the previously reported anti-flu mechanism of PLSCR1 also relies on its nuclear localization signal to restrict the import of IAV NP[20]. Furthermore, besides *IFN-λR1*, PLSCR1 enhances the expression of a select subset of ISGs in IAV infection as well, a mechanism potentially mediated by nuclear PLSCR1 on ISG gene transcription[23]. On the other hand, membrane PLSCR1 may modulate the JAK/STAT signaling pathway, thereby augmenting the optimal anti-viral activity of these ISGs[23]. We found that membrane PLSCR1 interacts with *IFN-λR1* protein in IAV infection, suggesting that it could facilitate viral elimination to some extent.

PLSCR1 is most well-known for its scramblase activity that favors PS exposure, apoptosis and phagocytosis[41, 46]. Using an enzymatically inactive mutant of PLSCR1, we uncoupled its lipid scramblase activity from anti-influenza activity. There are several potential explanations for this finding. First, our epithelial cell culture lacked phagocytes, therefore the impact of apoptosis followed by phagocytosis induced by PLSCR1 is minimal. Future studies using mice that harbor *Plscr1*(F281A) mutation would be needed to verify the role of lipid scramblase activity and epithelial cell apoptosis in the presence of phagocytes. Second, PLSCR1 exhibits only weak enzymatic activities compared to other members of lipid scramblase family, possibly due to its vastly different central β-barrel structure[15, 47]. PS externalization may be compensated by other more potent scramblases. Importantly, the lipid scramblase activity of PLSCR1 has been shown to be dispensable for its anti-SARS-CoV-2 function in a similar manner[15], suggesting a general lack of significance for its enzymatic activity in viral infections.

Although PLSCR1 has several previously described anti-influenza functions, including interfering with viral nuclear import[20], regulating TLR9 signaling [22], and potentiating the expression of other ISGs[23], our studies have clarified the relative contribution of the type 3 IFN pathway to *Plscr1*-mediated anti-influenza immunity using *Plscr1*^{-/-}*Ifn-λr1*^{-/-} mice. We observed that *Ifn-λr1*^{-/-} mice were more susceptible to IAV infection than *Plscr1*^{-/-}, suggesting that the complete loss of *Ifn-λr1* results in worse protection than impaired *Ifn-λr1* upregulation alone. Moreover, the previously identified anti-IAV functions of *Plscr1* do not appear sufficient to compensate for the loss of *Ifn-λr1* signaling in *Ifn-λr1*^{-/-} mice. The absence of further disease exacerbation or increased viral titers in *Plscr1*^{-/-}*Ifn-λr1*^{-/-} mice compared to *Ifn-λr1*^{-/-} mice indicates that the anti-influenza activity of *Plscr1* is largely dependent on *Ifn-λr1*.

While scRNA-seq analysis revealed that endothelial cells express *Plscr1* most abundantly in the lung, they are not the major target of IAV infection, and IAV does not efficiently replicate in them[48]. Instead, airway epithelial cells are the frontline defense against respiratory pathogens, with ciliated epithelial cells being the only cell type that express α2,3-linked SA, the primary influenza virus receptor in the mouse airway[49]. Coincidentally, our scRNA-seq results

showed that ciliated epithelial cells not only had the highest aggregated expression of *Plscr1*, but also had the most significant increase in *Plscr1* expression in early IAV infection at 3 dpi. Experiments with *Plscr1^{flloxStop}Foxj1-Cre⁺* mice further supported ciliated epithelial cell-dependent protection against IAV, with improved immunity and viral clearance, and dampened immunopathology at as early as 3 dpi. These findings suggest that as a result of enhanced *Ifn-λr1* due to *Plscr1* overexpression, type 3 interferons were able to exert their advantages being the earliest produced interferon, mounting both antiviral and anti-inflammatory responses in ciliated epithelial cells. To further establish the causal relationship between *Plscr1* and *Ifn-λ* signaling in airway ciliated epithelial cells, future experiments should focus on specifically overexpressing *Plscr1* in ciliated epithelial cells on an *Ifn-λr1^{-/-}* background by breeding *Plscr1^{flloxStop}Foxj1-Cre⁺Ifn-λr1^{-/-}* mice. In addition, ciliated epithelial cells isolated from *Ifn-λr1^{-/-}* murine airways could be transduced with a *Plscr1* overexpression construct. We hypothesize that overexpression of *Plscr1* in ciliated epithelial cells would not be able to rescue susceptibility in *Ifn-λr1^{-/-}* mice or cells, as the *Plscr1^{-/-}Ifn-λr1^{-/-}* mouse model suggests that *Plscr1*'s *Ifn-λr1*-independent anti-influenza mechanisms are likely minor compared to its role in upregulating *Ifn-λr1*.

Taken together, our findings highlight the essential role of PLSCR1 in the regulation of IFN-λR1 transcription in nucleus and expression on plasma membrane, both *in vitro* and *in vivo*. These mechanisms are crucial for inhibiting viral spread, reducing inflammation, and enhancing overall host fitness during IAV infection. Further, we found that the enzymatic activity of PLSCR1 is dispensable for its anti-influenza function. Finally, ciliated airway epithelial cells are the primary cell type in the lung for mounting PLSCR1-mediated anti-influenza responses. The potential of PLSCR1 agonists that target ciliated airway epithelial cells as therapeutic treatments for influenza holds promise for future medical interventions. Moreover, our results have the potential to impact the classical yet evolving field of IFN signaling. Not only do these findings elucidate and expand our understanding of newly discovered IFN-λ signaling, but they also shed light on the specific cell types and conditions under which IFN-λ signaling is modulated. Given the significance of IFN-λ signaling in various infectious diseases, these insights may pave the way for innovative therapeutics approaches targeting corresponding regulatory molecules in the treatment of other microbial infections in addition to influenza.

Materials and methods

Viruses

Purified A/WSN/1933 and A/PR/8/1934 (H1N1) virus (IAV) was kindly provided by Dr. Amanda Jamieson, Brown University. MDCK cells were used for the preparation of virus stocks and for virus titration by plaque assay.

Cell culture

Calu-3 cells were kindly provided by Dr. Suchitra Kamle, Brown University. Cells were maintained in Eagle's Minimum Essential Medium (ATCC) with 10% fetal bovine serum (Gibco) and 1% penicillin-streptomycin (Sigma-Aldrich).

MDCK cells were kindly provided by Dr. Amanda Jamieson, Brown University. Cells were maintained in Dulbecco's Modified Eagle Medium (Genesee) with 10% fetal bovine serum and 1% penicillin-streptomycin.

PLSCR1^{-/-} A549 cells were kindly provided by Dr. John MacMicking, Yale University. 293T cells were kindly provided by Dr. Suchitra Kamle, Brown University. Both cells were maintained in Dulbecco's Modified Eagle Medium with 10% fetal bovine serum, 1% MEM non-essential amino acids (Gibco) and 1mM sodium pyruvate (Gibco).

Mice

Wild-type (*Wt*) mice on a C57BL/6J genetic background were purchased from the Jackson Laboratories. *Plscr1*^{tm1a(EUCOMM)Hmgu} mice (*Plscr1*^{-/-} mice on C57BL/6 background) were purchased from the International Mouse Phenotyping Consortium. Those mice have an FRT flanked *lacZ*/neomycin sequence-tagged *LoxP* insertion upstream of *Plscr1* exon 6 and 7, and another *LoxP* insertion downstream of these critical exons. Subsequent *Cre* expression results in whole-body knockout mice that transcribed a shortened and nonfunctional transcript of *Plscr1* [50]. Genetic screening of *Plscr1*^{-/-} mice was carried out by conventional PCR on genomic DNA from mouse tail biopsies using the following primers for *LacZ* reporter: Fw: 5'-GCGATCGTAATCACCCGAGT-3' and Rev: 5'-CCGCCAAGACTGTTACCCAT-3'. This set of primers generates a 307bp fragment in *Plscr1*^{-/-} mice.

Rosa26 locus targeted *Plscr1* conditional knock-in transgenic mice (*Rosa26-LoxP-STOP-LoxP-Plscr1* Tg; *Plscr1*^{flloxStop} mice on C57BL/6 background) were generated at Brown Mouse Transgenic and Gene Targeting Facility. They were bred with *LysM-Cre* mice purchased from the Jackson Laboratories and *Foxj1-Cre* mice gifted by Drs. Yong Zhang and Michael Holtzman at Washington University School of Medicine in St. Louis to generate *Plscr1*^{flloxStop} *LysM-Cre*⁺ and *Plscr1*^{flloxStop} *Foxj1-Cre*⁺ mice respectively. For the detection of *Cre* recombinase in *LysM-Cre* mice, the following primers were used: Common Fw: 5'-CTTGGGCTGCCAGAATTCTC-3', Mutant Rev: 5'-CCCAGAAATGCCAGATTACG-3' and Wild type Rev: 5'-TTACAGTCGGCCAGGCTGAC-3'. A 700bp fragment is amplified in homozygous mutants, while a 350bp fragment is amplified in wild type mice. Both fragments show up in heterozygotes. For the detection of *Cre* recombinase in *Foxj1-Cre* mice, the following primers were used: Mutant Fw: 5'-CGTATAGCCGAAATTGCCAGG-3' and Mutant Rev: 5'-CTGACCAGAGTCATCCTTAGC-3'. A 327bp fragment is amplified in homozygous mutants.

Ifn- λ r1^{tm1a(EUCOMM)Wtsi}; Deleter-Cre mice (*Ifn- λ r1*^{-/-} mice on C57BL/6 background) were generated and gifted by Dr. Sanghyun Lee at Brown University [51, 52]. They were bred with *Plscr1*^{-/-} mice to generate *Plscr1*^{-/-} *Ifn- λ r1*^{-/-} mice. For the detection of *Ifn- λ r1* in those mice, the following primers were used: Common Fw: 5'-AGGGAAGCCAAGGGGATGGC-3'; Rev 1: 5'-AGTGCTGCTGAGGACCAGGA-3'; Rev 2: 5'-GGCTCTGGACCTACGCGCTG-3'. A 564bp fragment is amplified in homozygous *Ifn- λ r1*^{-/-} mutants, while a 231bp fragment is amplified in wild type mice.

All mice were housed and further bred in Brown University animal facilities. The mice with null mutation or overexpression of *Plscr1* did not show any apparent abnormal phenotypes and developmental and signaling issues. All murine procedures were approved by the Institutional Animal Care and Use Committees at Brown University.

Infection and treatment of mice

10-12-week-old mice were intranasally infected once with various doses of IAV in 30 μ L of sterile PBS (Gibco), or treated with 2.5 μ g/g of body weight of poly(I:C) (Invivogen) every day for 6 days under isoflurane-based anesthesia. Poly(I:C) was warmed up in a water bath at 37 °C before administration. Thirty μ L of sterile PBS was administered in control mice. Only males were used in infections. Littermates were randomly assigned to experimental and control groups.

BAL harvest and differential cell counts

Mice were euthanized with intra-peritoneal injection of 300 μ L of urethane (0.18 g/ml, Sigma-Aldrich). Bronchoalveolar lavage (BAL) of the whole lung was performed with a total of 1 mL ice-cold PBS via an incision at the trachea. An aliquot was stained with trypan blue solution (Gibco) for viability and cell number determination using TC20 automated cell counter (Bio-Rad). Samples

were centrifuged at 2500 rpm for 5 min at 4°C. Cells were pelleted onto glass slides via cytospin centrifugation at 700 rpm for 7 min and stained with Hema 3 (Fisher Scientific). Neutrophils, macrophages, lymphocytes and eosinophils were counted under a light microscope.

RNA isolation and qPCR

Harvested right lungs were immediately snap-frozen in liquid nitrogen and stored at -80°C afterwards. Frozen lungs were homogenized in TRIzol Reagent (Invitrogen). Cells were washed with ice cold PBS once, directly lysed in the culture dish with TRIzol Reagent and collected using a cell scraper. RNA isolation was performed with RNeasy Mini Kit (Qiagen). 0.5 µg of the isolated RNA was used for cDNA synthesis with iScript cDNA Synthesis Kit (Bio-Rad). Real-time quantitative PCR was performed with iTaq Universal SYBR® Green Supermix (Bio-Rad). All primers used were listed in Table S1. Relative amounts of mRNA expression were normalized to the level of mouse *Gapdh* or human *GAPDH*.

<i>qRT-PCR primers</i>		
Gene	Primer	Sequence (5'-3')
IAV M gene	Fwd	CATGGAATGGCTAAAGACAAGACC
	Rev	CCATTAAGGGCATTTTGGACA
<i>Plscr1</i>	Fwd	GCCCAAGTTCACTCTCCAAA
	Rev	GAGCTCAAAGTCAATGTCGG
<i>Ifn-α</i>	Fwd	AGTGAGCTGACCCAGCAGAT
	Rev	AGACAGCCTTGCAGGTCATT
<i>Ifn-β</i>	Fwd	CCCTATGGAGATGACGGAGA
	Rev	ACCCAGTGCTGGAGAAATTG
<i>Ifn-γ</i>	Fwd	ATGGCTATTTCTGGCTGTTACT
	Rev	AATGACGCTTATGTTGTTGCTG
<i>Ifn-λ2 and Ifn-λ3</i>	Fwd	AGTGAAGCAAAGGATTG
	Rev	GAGATGAGGTGGGAAGT
<i>Ifn-λr1</i>	Fwd	GACGAGTACAGGCAGCTTCC
	Rev	AGCATTGACCCTTAGGATCTTCTC
<i>Gapdh</i>	Fwd	AGGTCGGTGTGAACGGATTG
	Rev	TGTAGACCATGTAGTTGAGGTCA
<i>PLSCR1</i>	Fwd	CTGACTTCTGAGAAGGTTGC
	Rev	GAATGCTGTCGGTGGATACTG
<i>IFN-λ1</i>	Fwd	ACATTGGCAGGTTCAAATCTC
	Rev	TGAGTGACTCTTCCAAGGC
<i>IFN-λR1</i>	Fwd	CAGTGTCCCGAAATACAGCAAG
	Rev	TGTGTCCAGAAAAGTCCAGGGC
<i>IFN-λR1</i> promoter	Fwd	CCGGCCTTGAAGTCTCCCT
	Rev	GCGCTCGAAACTCGCCC
<i>GAPDH</i>	Fwd	TCGTGGAAGGACTCATGACC
	Rev	TCCACCACCCTGTTGCTGTA

Table S1.

PCR Primer List

Immunohistology and immunofluorescence

For immunohistology, immediately after euthanasia, left lungs were inflated through the incision at the trachea with 10% buffered formalin and soaked in formalin at RT. They were then transferred to 70% ethanol prior to paraffin embedding, sectioning and staining with hematoxylin

and eosin (H&E) performed at Brown University Molecular Pathology Core. Lungs were scanned with a VS200 slide scanner (Evident).

For immunofluorescence of mouse lungs, unstained paraffin-embedded lung sections were rehydrated in xylene (Fisher Scientific) and decreasing concentrations of ethanol (Pharmco). They were then steamed in antigen retrieval buffer (Abcam) for 30 minutes and blocked with 1% normal goat serum (Abcam) for 30 minutes at RT. Tissues were stained overnight at 4°C with goat anti-mouse Ifn- λ r1 polyclonal antibody (1:300 dilution, Invitrogen, lot #WL3462592C), rabbit anti-mouse uteroglobin polyclonal antibody (1:300 dilution, Santa Cruz Biotechnology, lot #F0220), rabbit anti-mouse Spc polyclonal antibody (1:100 dilution, Santa Cruz Biotechnology, lot #C2007), mouse anti-mouse Foxj1 monoclonal antibody (1:300 dilution, Invitrogen, lot #2712325), and/or rabbit anti-mouse Plscr1 polyclonal antibody (1:300 dilution, Proteintech, no lot/clone #). Tissues were washed 3 times with PBS and stained with chicken anti-goat Alexa 594 (Invitrogen, lot #2318436), goat anti-rabbit Alexa 488 (Invitrogen, lot #2051237) and/or donkey anti-goat Alexa 405 (Invitrogen, lot #XI353690) for 1 hour at RT in the dark. For visualization of IAV, tissues were stained overnight at 4°C with goat anti-H1N1-FITC (1:500 dilution, US Biological, lot #L23091555 C23082906). After washing 3 times with PBS, sections were mounted with Vectashield® mounting medium with or without DAPI (Vector Laboratories). Lungs were visualized under an APEXVIEW APX100 microscope (Evident).

For immunofluorescence of cell cultures, cells were seeded in Millicell EZ slides (Millipore Sigma). After washing with ice-cold PBS, 4% paraformaldehyde (Boston BioProducts) was added on ice for 10 minutes for fixation and removed. If necessary, 0.2% Triton X-100 (Sigma-Aldrich) was added for 10 minutes to permeabilize the cells on ice, followed by 3 times of PBS wash. Cells were blocked with 5% bovine serum albumin (Fisher Scientifics) for 1 hour at RT. They were stained overnight at 4°C with mouse anti-human PLSCR1 monoclonal antibody (1:50 dilution, R&D Systems, clone #875327) and rabbit anti-human IFN- λ R1 polyclonal antibody (1:100 dilution, Invitrogen, lot #ZB4228832A). Cells were washed 3 times with PBS and stained with chicken anti-mouse Alexa 594 (Invitrogen, lot #2482957) and goat anti-rabbit Alexa 488 for 1 hour at RT in the dark. After washing 3 times with PBS, slides were mounted with Vectashield® mounting medium with DAPI and visualized under a Nikon ECLIPSE Ti microscope.

When necessary, fluorescent images were analyzed using ImageJ[53]. Areas of interest were selected using freehand selection. Area, mean grey value and integrated density were measured. For H1N1 staining, uninfected lungs were used as background readings. For Ifn- λ r1 staining, a region without fluorescence was used as a background reading. Corrected total cell fluorescence (CTCF) was calculated using the following formula: $CTCF = \text{Integrated Density} - (\text{Area of selection} \times \text{Mean fluorescence of background readings})$.

Proximity ligation assay (PLA)

Duolink® PLA fluorescence (Millipore Sigma) was performed according to manufacturer's protocol. In brief, unstained paraffin-embedded lung sections were rehydrated, retrieved and blocked with Duolink® Blocking Solution for 60 minutes at 37°C. They were then incubated with goat anti-mouse Ifn- λ r1 polyclonal antibody and rabbit anti-mouse Plscr1 polyclonal antibody overnight at 4°C. Next, slides were incubated with anti-goat PLUS and anti-rabbit MINUS probes for 1 hour at 37°C. For cell culture, A549 cells were seeded in Millicell EZ slides, fixed and permeabilized. After blocking, cells were incubated with rabbit anti-human IFN- λ R1 polyclonal antibody and mouse anti-human PLSCR1 monoclonal antibody overnight at 4°C. Next, cells were incubated with anti-mouse PLUS and anti-rabbit MINUS probes for 1 hour at 37°C. Diluted ligase and polymerase were applied to samples for 30 minutes at 37°C in order. Finally, slides were mounted with Duolink® In Situ Mounting Medium with DAPI and imaged with an APEXVIEW APX100 microscope.

IAV Infection and rhIFN- λ 1 treatment in cell culture

Calu-3 cells at 80-90% confluency were washed twice with Opti-MEM (Gibco) and infected with IAV (WSN) at 0.1 multiplicity of infection (MOI) in Opti-MEM. One hour after viral attachment at 4°C, IAV supernatant was removed. Cells were washed once and incubated in fresh Opti-MEM for 23 hours at 37°C.

A549 cells at 80-90% confluency were washed once with PBS and infected with IAV (PR8) at 1, 5 or 10 MOI in PBS supplemented with BSA, CaCl₂ and MgCl₂. One hour after viral attainment at 37°C, IAV supernatant was removed. Cells were incubated in Opti-MEM with 10 µg/mL trypsin for 23 hours at 37°C.

Calu-3 or A549 cells were treated with 10 or 100 ng/mL of recombinant human IL-29/IFN-lambda 1 Protein (R&D Systems) for 6 or 24 hours at 37°C. In control Calu-3 groups, cells were first incubated with 1 µg/mL of anti-human interferon lambda receptor 1 neutralizing antibody (PBL Assay Science, clone #MMHLR-1) for 1 hour, and then treated with IFN- λ .

Co-immunoprecipitation (Co-IP)

Cryopreserved mouse lungs were homogenized in Pierce RIPA buffer (Thermo Scientific). Protein concentrations were determined using Pierce BCA Protein Assay Kit (Thermo Scientific) according to the manufacturer's protocol. Catch and Release® v2.0 Reversible Immunoprecipitation System (Millipore) was used according to manufacturer's protocol. In brief, 500 µg of lung lysate, 2 µg of rabbit anti-human Plscr1 polyclonal antibody or negative control human IgG and 10 µL of antibody capture affinity ligand were added to the spin columns with resin slurry. The mixture was incubated on a rotator at 4°C overnight. The unbound flow-through was collected by centrifuge. Bound proteins were washed 3 times and eluted using denaturing elution buffer (5% SDS, 8M Urea, and 100mM Glycine).

Western blot

Cryopreserved mouse lungs were homogenized in Pierce RIPA buffer. Cells in culture were washed with ice-cold PBS, lysed in Pierce RIPA buffer for 10 minutes on ice, and then scraped into RIPA buffer. Protein concentrations were determined using Pierce BCA Protein Assay Kit according to the manufacturer's protocol. Twenty µg of each sample was loaded into 4-20% Mini-PROTEAN TGX Gels (Bio-Rad) and run for 1 hour 20 minutes at 125V in Tris/Glycine/SDS running buffer (Bio-Rad). Gels were transferred using Trans-Blot Turbo Transfer Pack (Bio-Rad). Membranes were incubated with rabbit anti-mouse/human Plscr1 polyclonal antibody (1:1000 dilution), rabbit anti-mouse/human Ifn- λ 1 polyclonal antibody (1:1000 dilution, ABclonal, lot #0094910201), and/or mouse anti-mouse/human β -actin monoclonal antibody (1:2500 dilution, Santa Cruz Biotechnology, lot #F1323) overnight at 4°C. After washing with TBST (Boston BioProducts), membranes were incubated with HRP-linked anti-rabbit IgG (1:5000 dilution, Cell Signaling, lot #30) or HRP-linked anti-mouse IgG (1:5000 dilution, Cell Signaling, lot #36) for 1.5 hour at RT. Finally, membranes were treated with SuperSignal West Pico PLUS Chemiluminescent Substrate (Thermo Scientific) and imaged with ChemiDoc Imaging Systems (Bio-Rad).

Chromatin-immunoprecipitation (CHIP)

SimpleCHIP® Enzymatic Chromatin IP (Cell Signaling) was performed according to manufacturer's protocol. In brief, IAV-infected Calu-3 cells were fixed with formaldehyde to cross-link proteins to DNA. Next, chromatin was digested with Micrococcal Nuclease into 150-900 bp DNA/protein fragments. Nuclear membrane was broken by sonication. A portion of chromatin preparation was purified for DNA prior to immunoprecipitation to confirm the digestion by electrophoresis and concentration by OD260. One µg rabbit anti-mouse Plscr1 polyclonal antibody, negative control normal rabbit IgG, or positive control Histone H3 rabbit monoclonal antibody

was added to 5µg chromatin preparation. The complex co-precipitated and was captured by Protein G magnetic beads. After 4 low salt washes and 2 high salt washes, chromatin was eluted and cross-links were reversed with proteinase K. DNA was purified using spin columns and analyzed by both standard PCR and qRT-PCR. For qRT-PCR, percent input method was used according to the following formula: Percent Input = $2\% \times 2^{(C[T] \text{ 2\%Input Sample} - C[T] \text{ IP Sample})}$.

Transformation of competent cells

All *PLSCR1* plasmids were provided by Dr. John MacMicking's Lab at Yale University. Plasmid DNA concentrations were determined using a spectrophotometer. Fifty µl of DH5α competent cells (Invitrogen) and 50ng of each plasmid was mixed and incubated for 30 minutes on ice. Then, the plasmid and competent cell mixtures were heat shocked for 45 seconds at 42°C, and cooled on ice for 2 minutes. Super Optimal broth with Catabolite repression (SOC) media (homemade with 2% tryptone, 0.5% yeast extract, 10mM NaCl, 2.5mM KCl, 10 mM MgCl₂, 10 mM MgSO₄, and 20 mM glucose) was added to bacteria. The cells were grown in a shaking incubator at 200RPM for 45 minutes at 37°C. The mixtures were then plated on LB agar ampicillin plates and incubated at 37°C overnight.

Single colonies were selected for each plate, inoculated in Super Optimal Broth (SOB) media (homemade with 2% tryptone, 0.5% yeast extract, 10mM NaCl, 2.5mM KCl, 10 mM MgCl₂, and 10 mM MgSO₄) and grown to OD₆₀₀ of 0.6 at 37°C for 12-18 hours, with vigorous shaking of 200RPM. Cultures were then centrifuged at 6000G for 15 minutes at 4°C. Cell pellets were used for plasmid isolation.

DNA plasmid isolation

DNA plasmids were isolated using the QIAfilter Maxi Cartridges and QIAGEN Plasmid Maxi Kit, according to the manufacture's protocol. In brief, *PLSCR1*-transduced competent cell pellets were resuspended in buffer P1 containing RNase A solution and lyse blue. After adding Buffer P2, mixtures were incubated at RT for up to 5 minutes. Buffer P3 was then mixed with the lysates, transferred into the barrel of the QIAfilter cartridges and incubated at RT for 10 minutes. Cell lysates were filtered through the equilibrated QIAGEN-tip by gravity. DNA was eluted with Buffer QF and precipitated with isopropanol by centrifuging at 15000G for 10 minutes. Pellets were airdried for 10 minutes and dissolved in di-water. A spectrophotometer was used to determine plasmid concentrations.

Lentiviral packaging

293T cells were grown in DMEM until 90-95% confluency and then switched to Opti-MEM for 2 hours. Ten µg of plasmid of interest was mixed with lipofectamine 300 transfection reagent (Invitrogen), pPACKH1 HIV Lentivector Packaging Kit (System Biosciences) and Opti-MEM, and incubated for 20mins at RT. After removing half media from the cell culture dishes, DNA-lipid complex was added and cells were incubated at 37°C with 5% CO₂ for 6 hours. After 6 hours, culture supernatants were replaced with viral harvesting media (DMEM with glutamine, 10% FBS, 1% MEM non-essential amino acids, 1mM sodium pyruvate, 2% BSA, and 1:50 Hank's solution) for overnight incubation. Viral supernatant was collected every day for the next 4 days by centrifuging cell culture supernatant at 1200G for 5 minutes. Viral supernatant was mixed with lentiviral concentrator solution (40% polyethylene glycol 8000 and 1.2M NaCl in PBS) at 4:1, and centrifuged at 1600G for 60 minutes at 4°C. The viral pellets were resuspended with Opti-MEM and stored at -80°C.

Lentiviral transduction

PLSCR1^{-/-} A549 cells were seeded in a 96 well plate with DMEM. Lentiviral stocks were serially diluted in DMEM and polybrene (Millipore Sigma) mixture from 1:10 to 1:10,000, and added to cell culture. The 96 well plate was centrifuged at 2000RPM for 30 minutes at RT and incubated at 37°C

with 5% CO₂ overnight. Media was replaced with fresh DMEM without polybrene the next day. After 48 hours post transduction, DMEM with 200ug/ml hygromycin was added for antibiotic selection for 10 days.

Flow cytometry

Antibodies were titrated prior to confirming transduction efficiency by flow cytometry. One million cells were used for each sample. Samples were stained with LIVE/DEAD™ fixable violet dead cell stain for 405 nm excitation (1:1000, Invitrogen, lot #2581659) for 30 minutes. Surface and cytoplasm staining samples were fixed with 4% paraformaldehyde for 20 minutes at RT. Nuclear staining samples were fixed with true nuclear fix concentrate (BioLegend) for 45 minutes at RT. All samples were stained with mouse anti-human PLSCR1 monoclonal antibody (1:500) for 20 minutes, and chicken anti-mouse Alexa 594 (1:1000) for 20 minutes on ice. After fixation and each antibody incubation, surface staining samples were washed twice with FACS buffer (1% bovine serum albumin), cytoplasm staining samples were washed twice with Perm/Wash buffer (BD), and nuclear staining samples were washed twice with true nuclear perm buffer (BioLegend). Final samples were resuspended in FACS buffer. Flow cytometry analysis was performed with FACS Aria IIIu (BD) by Brown University Flow Cytometry and Sorting Facility.

Cell coverage assay

After IAV infection, A549 cells in 12-well plates were fixed with 4% paraformaldehyde in PBS for 1 hour at RT. Cells were then stained with 1% crystal violet solution (Sigma-Aldrich) for 5 minutes, washed with di-water 3 times and air-dried. Plates were scanned with an APEXVIEW APX100 microscope. Brightfield images were converted to 8-bit and the threshold was adjusted using a dark background in ImageJ[53]. Areas of interest were selected using oval selection. Cell coverages were quantified by measuring mean grey values of each well.

Plaque assay

For plaque assay of IAV titer in mouse lungs, a small piece of frozen lung was weighed and homogenized in 10mL decarbonated DMEM (MP Biomedicals) per 1g of lung. The homogenizer was sterilized in between samples with washes in the following order: twice with 0.1% SDS (Bio-Rad), 1:100 diluted bleach in di-water, once with 0.1% SDS, 0.001% Coomassie blue in di-water, twice with 70% ethanol, and once with sterile PBS. Homogenates were centrifuged at 2000G for 5 minutes at RT, and supernatant was collected.

For plaque assay of IAV titer in cell cultures, flu supernatant was collected, centrifuged at 2000G for 10 minutes, and filtered through 0.45µm filter.

MDCK cells were grown in 6-well plates to ~80% confluency. Virus was diluted to desired concentrations with PBS supplemented with BSA, CaCl₂ and MgCl₂. Media was aspirated from MDCK cell cultures and replaced with 100µL of viral samples in each well. The plates were incubated for 1 hour at 37°C and agitated every 15 minutes to prevent from drying. After 1 hour, flu was aspirated and the wells were filled with 2mL plaque assay overlay (DMEM/F-12 (Gibco) supplemented with NaHCO₃, BSA, DEAE dextran, and penicillin-streptomycin, mixed with 2% bacteriological agar (Oxoid) at 1:1 ratio). The plates were left at RT in the hood for 5 minutes to let the agar solidify, and then placed upside down in the 37°C incubator for 3 days until visible plaques were observed. Cells were fixed with 4% paraformaldehyde at RT for 1 hour. Agar overlay was removed with a metal spatula. Cells were stained with crystal violet solution gram (Harleco) for 5 minutes, and washed 3 times with di-water. Plates were air-dried. IAV titer was determined using the following formula: PFU/mL = # of plaques * 1/dilution factor * 10.

Bulk RNA sequencing

Selected RNA isolated from mouse lungs in the same treatment group were pooled together into one sample. Pooled RNA concentrations were measured using a Nanodrop (Thermo Scientific). Quality control, library preparation, bulk RNA sequencing and data analysis was performed by Azenta Genomics/GENEWIZ. Heatmaps were made with expressions of selected genes in Morpheus (<https://software.broadinstitute.org/morpheus>). Hierarchical clustering was performed with one minus pearson correlation, complete linkage method and clustering by row. ISGs were identified using the Interferome database, with selections of a fold change ≥ 2 in all cell types in lung of *Mus musculus*.

Single cell RNA sequencing analysis

Single cell RNA sequencing dataset were generated from uninfected or IAV-infected mouse lungs at 0, 1, 3, 6 and 21 dpi as previously described. Single-cell transcriptomic libraries were generated using the 5' Gene Expression Kit (V2, 10X Genomics) according to the manufacturer's instructions with the addition of primers to amplify HTOs during cDNA amplification. Sequencing was performed on the Illumina NovaSeq to generate approximately 500 million reads per sample. The 10X gene-expression data were processed and normalized using CellRanger (v.3.0.2, 10X Genomics). QC was performed first by excluding any gene that was not present in at least 0.1% of total called cells and then by excluding cells that exhibited extremes in the species-specific distributions of: the number of genes expressed (<100 or $>5,550$), the number of mRNA molecules ($>30,000$), or the percentage of expression owed to mitochondrial genes ($>7.5\%$). Scaling, principal component analysis, dimensionality reduction and cell clustering were performed using Seurat algorithm as previously described. Cell clusters were annotated using known markers from the literature (Table S2).

All following analysis were performed using R Statistical Software (v4.3.0). Two-dimensional UMAPs were generated with DimPlot (Seurat). Bar charts of proportion and cell count of each cluster were plotted with `geom_col` (ggplot2). Violin plots of *Plscr1* expression in different cell clusters were plotted with VlnPlot (Seurat). Differentially expressed genes for ciliated epithelial cell clusters at different timepoints in comparison to 0 dpi were identified with FindMarkers (Seurat). Those genes were used for Gene Ontology (GO) analysis with `enrichGo` (clusterProfiler), and heatmaps with DoHeatmap (Seurat).

<i>scRNA-seq cluster annotations</i>		
Cluster	Cell Type	Transcriptional Marker
0	AT2 Cells	<i>Epcam, Sftpc</i>
1	Mfap5+ Fibroblasts	<i>Pdgfra, Mfap5</i>
2	Microvascular Endothelial Cells	<i>Pecam1, Gpihbp1</i>
3	Damage-Responsive Fibroblasts	<i>Pdgfra, Wt1</i>
4	T Cells	<i>Ptprc, Cd3e</i>
5	Activated Microvascular Endothelial Cells	<i>Pecam1, Gpihbp1, S1pr1</i>
6	Airway Smooth Muscle Cells	<i>Hhip, Acta2</i>
7	IAV-Infected Epithelial Cells	<i>Epcam, Flu</i>
8	Microvascular Endothelial Cells	<i>Pecam1, Gpihbp1</i>
9	High Mitochondrial Content Cells	
10	Krt8+ Epithelial Cells	<i>Epcam, Krt8</i>

11	AT2 Cells	<i>Epcam, Sftpc</i>
12	Ciliated Epithelial Cells	<i>Epcam, Foxj</i>
13	Club Cells	<i>Epcam, Scgb3a2</i>
14	B Cells	<i>Ptprc, Cd19</i>
15	Myofibroblasts	<i>Pdgfra, Acta2</i>
16	Neutrophils	<i>Ptprc, Mmp9</i>
17	Alveolar Fibroblasts	<i>Pdgfra, Wnt2</i>
18	Epcam+Pecam1+ Cells	<i>Epcam, Pecam1</i>
19	Monocytes	<i>Ptprc, Ly6c2</i>
20	Interferon-Responsive Fibroblasts	<i>Pdgfra, Bst2</i>
21	Mesothelial Cells	<i>Msln, Wt1</i>
22	Epcam+Col1a2+ EMT Cells	<i>Epcam, Col1a2</i>
23	Interstitial Macrophages	<i>Ptprc, Cd68, C1qb</i>
24	Lymphatic Endothelial Cells	<i>Pecam1, Prox1</i>
25	Aerocytes	<i>Pecam 1, Car4</i>
26	Vascular Smooth Muscle Cells	<i>Acta2, Pdgfrb, Notch3</i>
27	AT1 Cells	<i>Epcam, Akap5, Ager</i>
28	Macrovascular Endothelial Cells	<i>Pecam1, Vwf</i>
29	Ciliated Epithelial Cells	<i>Epcam, Foxj1</i>
30	Alveolar Macrophages	<i>Ptprc, Cd68, Chil3</i>
31	NK Cells	<i>Ptprc, Klrb1c</i>
32	Epcam+Col1a2+ EMT Cells	<i>Epcam, Col1a2</i>
33	Immune Doublets	<i>Ptprc, Cd3e, Cd19</i>
34	Regulatory T cells	<i>Cd4, Il2ra</i>
35	Basophils/Mast Cells	<i>Ptprc, Fcer1a, Gata2</i>
36	Multiplatelets	
37	Platelets	

Table S2.

scRNA-Seq Cluster Annotations

Quantitation and statistical analysis

Data were analyzed on GraphPad Prism software. Logrank (Mantel-Cox) test was used to compare survival rates. Ordinary two-way ANOVA tests were used to compare weight losses. Statistical significance of differences was assessed using the parametric Student's two-tailed t tests for all other normally distributed data. Differences were considered significant when $p < 0.05$. Outlier tests with the ROUT method ($Q = 1\%$) were used to identify and remove any outliers.

Acknowledgements

The authors thank A. Ayala, A. Jamieson and C. Lee for advice; J. MacMicking for sharing *PLSCR1*^{-/-} and *T16F*^{-/-} A549 cells and all *PLSCR1* plasmids; Brown University Molecular Pathology Core for histopathology; and Brown University Flow Cytometry and Sorting Facility for flow cytometry.

This work was supported by grants R01 HL146498 (YZ), P20 GM103652 (YZ), U54 GM115677 (YZ), T32 HL134625 (PS), AHA 24TPA1277918 (YZ), ATS 23-24PHP11 (YZ), R01 AI183155 (SL), and R35 HL145242 (MJH).

During the preparation of this work the author(s) used ChatGPT developed by OpenAI in order to check grammar and improve readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Additional information

Data availability

Sequencing data have been deposited in GEO under accession codes GSE307868.

Funding

National Heart, Lung, and Blood Institute (R01 HL146498)

- Yang Zhou

National Institute of General Medical Sciences (P20 GM103652)

- Yang Zhou

National Institute of General Medical Sciences (U54 GM115677)

- Yang Zhou

National Heart, Lung, and Blood Institute (T32 HL134625)

- Parand Sorkhdini

American Heart Association

<https://doi.org/10.58275/AHA.24TPA1277918.pc.gr.196624> 

- Yang Zhou

American Thoracic Society (23-24PHP11)

- Yang Zhou

National Heart, Lung, and Blood Institute (R35 HL145242)

- Michael Holtzman

National Institute of Allergy and Infectious Diseases (R01 AI183155)

- Sanghyun Lee

Additional files

Supplemental Figures [↗](#)

References

1. Holbach M (2016) **[Burden of disease from influenza-like illness among hospital staff and effectiveness of influenza vaccination during the flu epidemic at the beginning of 2015]** *MMW Fortschr Med* **158**:8–13 [Google Scholar](#)
2. (2020) **Antiviral drugs for influenza** *Med Lett Drugs Ther* **62**:1–4 [Google Scholar](#)
3. Sheppard P., et al. (2003) **IL-28, IL-29 and their class II cytokine receptor IL-28R** *Nat Immunol* **4**:63–8 [Google Scholar](#)
4. Donnelly R.P., et al. (2004) **The expanded family of class II cytokines that share the IL-10 receptor-2 (IL-10R2) chain** *J Leukoc Biol* **76**:314–21 [Google Scholar](#)
5. Swiecki M., Colonna M. (2011) **Type I interferons: diversity of sources, production pathways and effects on immune responses** *Curr Opin Virol* **1**:463–75 [Google Scholar](#)
6. Coccia E.M., et al. (2004) **Viral infection and Toll-like receptor agonists induce a differential expression of type I and lambda interferons in human plasmacytoid and monocyte-derived dendritic cells** *Eur J Immunol* **34**:796–805 [Google Scholar](#)
7. Sommereyns C., et al. (2008) **IFN-lambda (IFN-lambda) is expressed in a tissue-dependent fashion and primarily acts on epithelial cells in vivo** *PLoS Pathog* **4**:e1000017 [Google Scholar](#)
8. Broggi A., et al. (2017) **IFN-lambda suppresses intestinal inflammation by non-translational regulation of neutrophil function** *Nat Immunol* **18**:1084–1093 [Google Scholar](#)
9. Yin Z., et al. (2012) **Type III IFNs are produced by and stimulate human plasmacytoid dendritic cells** *J Immunol* **189**:2735–45 [Google Scholar](#)

10. Galani I.E., et al. (2017) **Interferon-lambda Mediates Non-redundant Front-Line Antiviral Protection against Influenza Virus Infection without Compromising Host Fitness** *Immunity* **46**:875–890 [Google Scholar](#)
11. Klinkhammer J., et al. (2018) **IFN-lambda prevents influenza virus spread from the upper airways to the lungs and limits virus transmission** *eLife* **7** [Google Scholar](#)
12. Chen X., et al. (2018) **Host Immune Response to Influenza A Virus Infection** *Front Immunol* **9**:320 [Google Scholar](#)
13. Zhou Q., et al. (2000) **Transcriptional control of the human plasma membrane phospholipid scramblase 1 gene is mediated by interferon-alpha** *Blood* **95**:2593–9 [Google Scholar](#)
14. Lu B., et al. (2007) **Expression of the phospholipid scramblase (PLSCR) gene family during the acute phase response** *Biochim Biophys Acta* **1771**:1177–85 [Google Scholar](#)
15. Xu D., et al. (2023) **PLSCR1 is a cell-autonomous defence factor against SARS-CoV-2 infection** *Nature* **619**:819–827 [Google Scholar](#)
16. Zhou Q., et al. (2005) **Phospholipid scramblase 1 binds to the promoter region of the inositol 1,4,5-triphosphate receptor type 1 gene to enhance its expression** *J Biol Chem* **280**:35062–8 [Google Scholar](#)
17. Huang P., et al. (2020) **Nuclear translocation of PLSCR1 activates STAT1 signaling in basal-like breast cancer** *Theranostics* **10**:4644–4658 [Google Scholar](#)
18. Karlsson M., et al. (2021) **A single-cell type transcriptomics map of human tissues** *Sci Adv* **7** [Google Scholar](#)
19. Basse F., et al. (1996) **Isolation of an erythrocyte membrane protein that mediates Ca²⁺-dependent transbilayer movement of phospholipid** *Journal of Biological Chemistry* **271**:17205–17210 [Google Scholar](#)
20. Luo W., et al. (2018) **Phospholipid scramblase 1 interacts with influenza A virus NP, impairing its nuclear import and thereby suppressing virus replication** *PLoS Pathog* **14**:e1006851 [Google Scholar](#)
21. Liu Y., et al. (2022) **ILDR1 promotes influenza A virus replication through binding to PLSCR1** *Sci Rep* **12**:8515 [Google Scholar](#)
22. Talukder A.H., et al. (2012) **Phospholipid scramblase 1 regulates Toll-like receptor 9-mediated type I interferon production in plasmacytoid dendritic cells** *Cell Res* **22**:1129–39 [Google Scholar](#)
23. Dong B., et al. (2004) **Phospholipid scramblase 1 potentiates the antiviral activity of interferon** *J Virol* **78**:8983–93 [Google Scholar](#)
24. Brandes M., et al. (2013) **A systems analysis identifies a feedforward inflammatory circuit leading to lethal influenza infection** *Cell* **154**:197–212 [Google Scholar](#)
25. Rusinova I., et al. (2013) **Interferome v2.0: an updated database of annotated interferon-regulated genes** *Nucleic Acids Res* **41** (Database issue):D1040–6 [Google Scholar](#)

26. Wiedmer T., et al. (2003) **Palmitoylation of phospholipid scramblase 1 controls its distribution between nucleus and plasma membrane** *Biochemistry* **42**:1227–1233 [Google Scholar](#)
27. Chen M.H., et al. (2005) **Phospholipid scramblase 1 contains a nonclassical nuclear localization signal with unique binding site in importin alpha** *J Biol Chem* **280**:10599–606 [Google Scholar](#)
28. Boyd D.F., et al. (2020) **Exuberant fibroblast activity compromises lung function via ADAMTS4** *Nature* **587**:466–471 [Google Scholar](#)
29. Meixner A., et al. (2010) **Jun and JunD-dependent functions in cell proliferation and stress response** *Cell Death Differ* **17**:1409–19 [Google Scholar](#)
30. Kraja A.T., et al. (2019) **Associations of Mitochondrial and Nuclear Mitochondrial Variants and Genes with Seven Metabolic Traits** *Am J Hum Genet* **104**:112–138 [Google Scholar](#)
31. Borges K.S., Arboleda V.A., Vilain E. (2015) **Mutations in the PCNA-binding site of CDKN1C inhibit cell proliferation by impairing the entry into S phase** *Cell Div* **10**:2 [Google Scholar](#)
32. Hendershot L.M., et al. (1994) **Localization of the gene encoding human BiP/GRP78, the endoplasmic reticulum cognate of the HSP70 family, to chromosome 9q34** *Genomics* **20**:281–4 [Google Scholar](#)
33. Mattoo R.U.H., et al. (2013) **Hsp110 is a bona fide chaperone using ATP to unfold stable misfolded polypeptides and reciprocally collaborate with Hsp70 to solubilize protein aggregates** *J Biol Chem* **288**:21399–21411 [Google Scholar](#)
34. Humphreys D.T., et al. (1999) **Clusterin has chaperone-like activity similar to that of small heat shock proteins** *J Biol Chem* **274**:6875–81 [Google Scholar](#)
35. Andreakos E., Zanoni I., Galani I.E. (2019) **Lambda interferons come to light: dual function cytokines mediating antiviral immunity and damage control** *Curr Opin Immunol* **56**:67–75 [Google Scholar](#)
36. Lizak M., Yarovsky T.O. (2012) **Phospholipid scramblase 1 mediates type I interferon-induced protection against staphylococcal alpha-toxin** *Cell Host Microbe* **11**:70–80 [Google Scholar](#)
37. Chen Y., et al. (2013) **Wogonoside induces cell cycle arrest and differentiation by affecting expression and subcellular localization of PLSCR1 in AML cells** *Blood* **121**:3682–91 [Google Scholar](#)
38. Wyles J.P., et al. (2007) **Nuclear interactions of topoisomerase II alpha and beta with phospholipid scramblase 1** *Nucleic Acids Res* **35**:4076–85 [Google Scholar](#)
39. Huang H., et al. (2015) **Phospholipid Scramblase 1 Interacts with Midkine and Regulates Hepatic Cancer Cell Proliferation and Migration** *Clin Lab* **61**:1501–8 [Google Scholar](#)
40. Ding S., et al. (2014) **Epigenetic reprogramming of the type III interferon response potentiates antiviral activity and suppresses tumor growth** *PLoS Biol* **12**:e1001758 [Google Scholar](#)

41. Guo J., et al. (2020) **Transient Receptor Potential Canonical 5-Scramblase Signaling Complex Mediates Neuronal Phosphatidylserine Externalization and Apoptosis** *Cells* **9** [Google Scholar](#)
42. Sun J., et al. (2002) **Plasma membrane phospholipid scramblase 1 is enriched in lipid rafts and interacts with the epidermal growth factor receptor** *Biochemistry* **41**:6338–45 [Google Scholar](#)
43. Li Y., et al. (2006) **The negative c-Myc target onzin affects proliferation and apoptosis via its obligate interaction with phospholipid scramblase 1** *Mol Cell Biol* **26**:3401–13 [Google Scholar](#)
44. Amir-Moazami O., et al. (2008) **Phospholipid scramblase 1 modulates a selected set of IgE receptor-mediated mast cell responses through LAT-dependent pathway** *J Biol Chem* **283**:25514–25523 [Google Scholar](#)
45. Kametaka S., et al. (2003) **Identification of phospholipid scramblase 1 as a novel interacting molecule with beta -secretase (beta -site amyloid precursor protein (APP) cleaving enzyme (BACE))** *J Biol Chem* **278**:15239–45 [Google Scholar](#)
46. Zhao J., et al. (1998) **Level of expression of phospholipid scramblase regulates induced movement of phosphatidylserine to the cell surface** *J Biol Chem* **273**:6603–6 [Google Scholar](#)
47. Tang D., et al. (2022) **Scramblases and virus infection** *Bioessays* **44**:e2100261 [Google Scholar](#)
48. Han T., et al. (2021) **Influenza A virus infects pulmonary microvascular endothelial cells leading to microvascular leakage and release of pro-inflammatory cytokines** *PeerJ* **9**:e11892 [Google Scholar](#)
49. Ibricevic A., et al. (2006) **Influenza virus receptor specificity and cell tropism in mouse and human airway epithelial cells** *J Virol* **80**:7469–80 [Google Scholar](#)
50. Bult C.J., et al. (2019) **Mouse Genome Database (MGD) 2019** *Nucleic Acids Research* **47**:D801–D806 [Google Scholar](#)
51. Baldridge M.T., et al. (2017) **Expression of Ifnlr1 on Intestinal Epithelial Cells Is Critical to the Antiviral Effects of Interferon Lambda against Norovirus and Reovirus** *J Virol* **91** [Google Scholar](#)
52. Schwenk F., Baron U., Rajewsky K. (1995) **A cre-transgenic mouse strain for the ubiquitous deletion of loxP-flanked gene segments including deletion in germ cells** *Nucleic Acids Res* **23**:5080–1 [Google Scholar](#)
53. Schneider C.A., Rasband W.S., Eliceiri K.W. (2012) **NIH Image to ImageJ: 25 years of image analysis** *Nat Methods* **9**:671–5 [Google Scholar](#)
54. Hao Y., et al. (2021) **Integrated analysis of multimodal single-cell data** *Cell* **184**:3573–3587 [Google Scholar](#)
55. RCoreTeam (2023) **R: A Language and Environment for Statistical Computing** Vienna, Austria: R Foundation for Statistical Computing [Google Scholar](#)
56. Wickham H. (2016) **ggplot2: Elegant Graphics for Data Analysis** Springer-Verlag New York [Google Scholar](#)

57. Wu T., et al. (2021) **clusterProfiler 4.0: A universal enrichment tool for interpreting omics data** *Innovation (Camb)* 2:100141 [Google Scholar](#)

Author information

Alina X Yang

Department of Molecular Microbiology and Immunology, Brown University, Providence, United States

Lisa Ramos-Rodriguez

Department of Molecular Microbiology and Immunology, Brown University, Providence, United States

Parand Sorkhdini

Department of Molecular Microbiology and Immunology, Brown University, Providence, United States

Dongqin Yang

Department of Molecular Microbiology and Immunology, Brown University, Providence, United States

Carmelissa Norbrun

Department of Molecular Microbiology and Immunology, Brown University, Providence, United States

Sonoor Majid

Department of Molecular Microbiology and Immunology, Brown University, Providence, United States

Sanghyun Lee

Department of Molecular Microbiology and Immunology, Brown University, Providence, United States

Yong Zhang

Division of Pulmonary and Critical Care Medicine, Washington University School of Medicine in St. Louis, St Louis, United States

Michael J Holtzman

Division of Pulmonary and Critical Care Medicine, Washington University School of Medicine in St. Louis, St Louis, United States

David F Boyd

Department of Molecular, Cell and Developmental Biology, University of California, Santa Cruz, Santa Cruz, United States

Yang Zhou

Department of Molecular Microbiology and Immunology, Brown University, Providence, United States

ORCID iD: [0000-0001-6867-6443](#)

For correspondence: yang_zhou@brown.edu

Editors

Reviewing Editor

Ke Cheng

Yongchuan Hospital of Chongqing Medical University, Chongqing, China

Senior Editor

Carla Rothlin

Yale University, New Haven, United States of America

Reviewer #1 (Public review):

This manuscript by Yang et al. presents a potentially novel mechanism by which Plscr1 defends against influenza virus infection. Using a global knockout (KO) and a tissue-specific overexpression mouse model, the authors demonstrate that Plscr1-KO mice exhibit increased susceptibility and inflammation following IAV infection. In contrast, overexpression of Plscr1 in ciliated epithelial cells protects mice from infection. Through transcriptomic analysis in mice and mechanistic studies in cell culture models, the authors reveal that Plscr1 transcriptionally upregulates Ifnlr1 expression and physically interacts with this receptor on the plasma membrane, thereby enhancing IFN- λ -mediated viral clearance.

Overall, it's a well-performed study, however, causality between Plscr1 and Ifnlr1 expression needs to be more firmly established. This is because two recent studies of PLSCR1 KO cells infected with different viruses found no major differences in gene expression levels compared with their WT controls (Xu et al. Nature, 2023; LePen et al. PLoS Biol, 2024). There were also defects in the expression of other cytokines (type I and II IFNs plus TNF-alpha) so a clear explanation of why Ifnlr1 was chosen should also be given.

While Plscr1 has long been recognized as a cell-intrinsic antiviral restriction factor, few studies have explored its broader physiological role. This study thus provides interesting insights into a specific function of Plscr1 in IAV-permissive airway epithelial cells and its contribution to whole body anti-viral immunity.

Comments on revisions:

Most of the requested changes and experiments have been done. One very informative experiment is the expression of Plscr1 in Ifnlr1-KO cells to determine if it still inhibits IAV infection. The authors have indicated that this experiment is currently being pursued by crossing mice to introduce Plscr1 expression into ciliated epithelial cells on an Ifnlr1 KO background. It will show if there are Ifnlr1-independent anti-flu activities that still require Plscr1.

<https://doi.org/10.7554/eLife.104359.2.sa1>

Author response:

The following is the authors' response to the original reviews

Reviewer #1 (Public Review):

Overall, it's a well-performed study, however, causality between Plscr1 and Ifnlr1 expression needs to be more firmly established. This is because two recent studies of PLSCR1 KO cells infected with different viruses found no major differences in gene expression levels compared with their WT controls (Xu et al. Nature, 2023; LePen et al.

PLoS Biol, 2024). There were also defects in the expression of other cytokines (type I and II IFNs plus TNF-alpha) so a clear explanation of why *Ifnlr1* was chosen should also be given.

We appreciate the reviewer's reference to the two recently published research on PLSCR1's role in SARS-CoV-2 infections. We have also discussed those studies in the Introduction and Discussion sections of this manuscript. Here, we would like to clarify ourselves for the rationale of investigating *Ifn-λr1* signaling.

The reviewer mentioned "defects in the expression of other cytokines (type I and II IFNs plus TNF-alpha)" and requested a clearer explanation of why *Ifnlr1* was chosen for study. In our investigation of IAV infection, we observed no defects in the expression of type I and II IFNs or TNF-α in *Plscr1*^{-/-} mice; rather, these cytokines were expressed at even higher levels compared to WT controls (Figures 2D and 3A). This indicates that the type I and II IFN and TNF-α signaling pathways remain intact and are not negatively affected by the loss of *Plscr1*. Notably, *Ifn-λr1* expression is the only one among all IFNs and their receptors that is significantly impaired in *Plscr1*^{-/-} mice (Figure 3A), justifying our focused investigation of this receptor. To further clarify this point, we have expanded the explanation under the section titled "Plscr1 Binds to *Ifn-λr1* Promoter and Activates *Ifn-λr1* Transcription in IAV Infection" within the Results. The reviewer noted that previously published studies "found no major differences in gene expression levels compared with their WT controls", but neither study examined *Ifn-λr1* expression.

(1) The authors propose that Plscr1 restricts IAV infection by regulating the type III IFN signaling pathway. While the data show a positive correlation between Ifnlr1 and Plscr1 levels in both mouse and cell culture models, additional evidence is needed to establish causality between the impaired type III IFN pathway, and the increased susceptibility observed in Plscr1-KO mice. To strengthen this conclusion, the following experiments could be undertaken: (i) Measure IAV titers in WT, Plscr1-KO, Ifnlr1-KO, and Plscr1/ Ifnlr1-double KO cells. If the antiviral activity of Plscr1 is highly dependent on Ifnlr1, there should be no further increase in IAV titers in double KO cells compared to single KO cells; (ii) over-express Plscr1 in Ifnlr1-KO cells to determine if it still inhibits IAV infection. If Plscr1's main action is to upregulate Ifnlr1, then it should not be able to rescue susceptibility since Ifnlr1 cannot be expressed in the KO background. If Plscr1 over-expression rescues viral susceptibility, then there are Ifnlr1-independent mechanisms involved. These experiments should help clarify the relative contribution of the type III IFN pathway to Plscr1-mediated antiviral immunity.

We agree with the reviewer that additional evidence is necessary to establish causality between the impaired type III IFN pathway and the increased susceptibility observed in *Plscr1*-KO mice. As requested by the reviewer, and one step further, we have measured IAV titers in *Wt*, *Plscr1*^{-/-}, *Ifn-λr1*^{-/-}, and *Plscr1*^{-/-}*Ifn-λr1*^{-/-} mouse lungs, which provided us with more comprehensive information at the tissue and organismal level compared to cell culture models. Our results are detailed under "The Anti-Influenza Activity of *Plscr1* Is Highly Dependent on *Ifn-λr1*" within "Results" section and in Supplemental Figure 5. Importantly, there was no further increase in weight loss (Supplemental Figure 5B), total BAL cell counts (Supplemental Figure 5C), neutrophil percentages (Supplemental Figure 5D), and IAV titers (Supplemental Figure 5E) in *Plscr1*^{-/-}*Ifn-λr1*^{-/-} mouse lungs compared to *Ifn-λr1*^{-/-} mouse lungs. These findings indicate that the antiviral activity of *Plscr1* is largely dependent on *Ifn-λr1*.

We agree that overexpression of *Plscr1* on an *Ifn-λr1*^{-/-} background would provide additional evidence to support our conclusion from the *Plscr1*^{-/-}*Ifn-λr1*^{-/-} mice. In future studies, we plan to specifically overexpress *Plscr1* in ciliated epithelial cells on the *Ifn-λr1*^{-/-} background by breeding *Plscr1*^{floxStop}*Foxj1-Cre*⁺*Ifn-λr1*^{-/-} mice. In addition, ciliated epithelial cells isolated from *Ifn-λr1*^{-/-} murine airways could be transduced with a *Plscr1* construct for

overexpression. We hypothesize that overexpression of *Plscr1* in ciliated epithelial cells will not rescue susceptibility in *Ifn-λr1*^{-/-} mice or cells, since our *Plscr1*^{-/-}*Ifn-λr1*^{-/-} mouse model suggest that *Ifn-λr1*-independent anti-influenza functions of *Plscr1* are likely minor compared to its role in upregulating *Ifn-λr1*. These future plans have been added to the “Discussion” section, and we look forward to presenting our results in a forthcoming publication.

(3) In Figure 4, the authors demonstrate the interaction between Plscr1 and Ifnlr1. They suggest that this interaction modulates IFN-λ signaling. However, Figures 5C-E show that the 5CA mutant, which lacks surface localization and the ability to bind Ifnlr1, exhibits similar anti-flu activity to WT Plscr1. Does this mean the interaction between Plscr1 and Ifnlr1 is dispensable for Plscr1-mediated antiviral function? Can the authors compare the activation of IFN-λ signaling pathway in Plscr1-KO cells expressing empty vector, WT Plscr1, and 5CA mutant? This could be done by measuring downstream ISG expression or using an ISRE-luciferase reporter assay upon IFN-λ treatment.

We agree with the reviewer that downstream activation of the IFN-λ signaling pathway is a critical component of the proposed regulatory role of PLSCR1. As suggested, we attempted to perform an ISRE-luciferase reporter assay following IFN-λ treatment in PLSCR1 rescue cell lines by transfecting the cells with hGAPDH-rLuc (Addgene #82479) and pGL4.45 [luc2P/ISRE/Hygro] (Promega #E4041).

Despite extensive efforts over several months, we were unable to achieve expression of pGL4.45 [luc2P/ISRE/Hygro] in PLSCR1 rescue cells using either Lipofectamine 3000 or electroporation, as no firefly luciferase activity was detected at baseline or following IFN-λ treatment. In contrast, hGAPDH-rLuc was robustly expressed in these cells.

The pGL4.45 [luc2P/ISRE/Hygro] plasmid was obtained directly from Promega as a purified product, and its sequence was confirmed via whole plasmid sequencing. Additionally, both hGAPDH-rLuc and pGL4.45 [luc2P/ISRE/Hygro] were successfully expressed in 293T cells, indicating that neither the plasmids nor the transfection protocols are inherently faulty.

We suspect that prior modifications to the PLSCR1 rescue cells—such as CRISPR-mediated knockout and lentiviral transduction—may interfere with successful transfection of pGL4.45 [luc2P/ISRE/Hygro] through an as-yet-unknown mechanism. Although these results are disappointing, we will continue troubleshooting and plan to communicate in a separate manuscript once the luciferase assay is successfully established.

Reviewer #1 (Recommendations):

(1) In the introduction, the linkage between the paragraph discussing type III IFN and PLSCR1 needs to be better established. The mention of PLSCR1 being an ISG at the outset may help connect these two paragraphs and make the text appear more logical.

We apologize for the lack of linkage and logic between type 3 IFN and PLSCR1. We have introduced PLSCR1 as an ISG at the beginning of its paragraph as recommended.

(2) The statement that, “Intriguingly, PLSCR1 is also an antiviral ISG, as its expression can be highly induced by type 1 and 2 interferons in various viral infections[15, 16]. However, whether its expression can be similarly induced by type 3 interferon has not been studied yet.” is incorrect. Xu et al. tested the role of PLSCR1 in type III IFN-induced control of SARS-CoV-2 (ref. 24). This needs to be revised.

We apologize for the incorrect information in the introduction and have revised the paragraph with the proper citation.

(3) In Figure 3B, can the authors provide a comprehensive heatmap that includes all ISGs above the threshold, rather than only a subset? This would offer a more complete overview of the changes in type I, II, and III IFN pathways in *Plscr1*-KO mice.

As suggested by the reviewer, we have provided a comprehensive heatmap that includes all ISGs above the threshold in Figure 3C (previously Figure 3B). We identified a total of 1,113 ISGs in our dataset with a fold change ≥ 2 . Enlarged heatmaps with gene names are provided in Supplemental Figure 1. Among those ISGs, 584 are regulated exclusively by type 1 IFNs, and 488 are regulated by both type 1 and type 2 interferons. Unfortunately, the Interferome database does not include information on type 3 IFN-inducible genes in mice[1]. Although many ISGs were robustly upregulated in *Plscr1*^{-/-} infected lungs, consistent with inflammation data, a large subset of ISGs failed to be transcribed when *Ifn- λ r1* function was impaired, especially at 7 dpi. We suspect that those non-transcribed ISGs in *Plscr1*^{-/-} mice may be specifically regulated by type 3 IFN and represent interesting targets for future research. These results have been added to “*Plscr1* Binds to *Ifn- λ r1* Promoter and Activates *Ifn- λ r1* Transcription in IAV Infection” within “Results” section.

(4) In Figure 3C, 5B and 7H, immunoblots should also be included to measure changes of *Ifnlr1*/IFNLR1 protein level.

As requested by the reviewer, we have provided western blots measuring *Ifn- λ r1*/IFN- λ R1 protein level in Figure 5B and 7I. The protein expressions were consistent with the PCR results.

(5) In Figure 3H, the amount of RPL30 is also low in the anti-PLSCR1-treated and IgG samples, making it difficult to estimate if ChIP binding is genuinely impacted.

RPL30 Exon 3 serves as a negative control in the ChIP experiment and is not expected to bind either the anti-PLSCR1-treated or the IgG control samples. Anti-Histone H3 treatment is a positive control, with the treated sample expected to show binding to RPL30 Exon 3. We hope this clarification has addressed any further potential confusion from the reviewer.

(6) In Figure 4A, can the authors show a larger slice of the gel with molecular weight markers for both *Plscr1* and *Ifnlr1*. In the coIP, the binding may be indirect through intermediate partners. Proximity ligation assay is a more direct assay for interaction and can be stated as such.

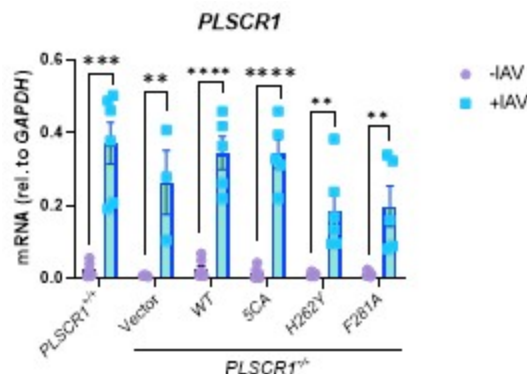
As suggested by the reviewer, we have included whole gel images of Figure 4A with molecular weight markers for both *Plscr1* and *Ifnlr1* in Supplemental Figure 3. We appreciate the reviewer’s affirmation of proximity ligation assay and have stated it as a more direct assay for interaction under “*Plscr1* Interacts with *Ifn- λ r1* on Pulmonary Epithelial Cell Membrane in IAV Infection” in “Results” section.

(7) In Figure 5A, how is the expression of PLSCR1 WT and mutants driven by an EF-1 α promoter can be further upregulated by IAV infection? Can the authors also use immunoblots to examine the protein level of PLSCR1?

We apologize for the confusion and appreciate the reviewer’s careful observation. We were initially surprised by this finding as well, but upon further investigation, we found out that the human *PLSCR1* primers used in our qRT-PCR assay can still detect the transcription from the undisturbed portion of the endogenous *PLSCR1* mRNA, even in *PLSCR1*^{-/-} cells. In the original Figure 5A, data for vector-transduced *PLSCR1*^{-/-} were not included because PCR was not performed on those samples at the time. After conducting PCR for vector-transduced

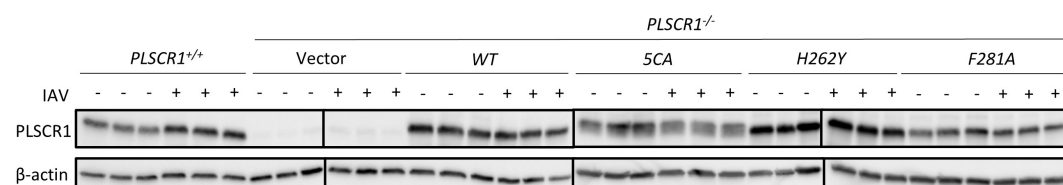
PLSCR1^{-/-} cells, we detected transcription of *PLSCR1*, which confirms that the signaling originates from endogenous DNA, but not from the EF-1α promoter-driven *PLSCR1* plasmid. Please see Author response image 1 below.

Author response image 1.



The forward human *PLSCR1* primer we used matches 15-34 nt of *Wt PLSCR1*, and the reverse primer matches 224-244 nt of *Wt PLSCR1*. CRISPR-Cas9 KO of *PLSCR1* was mediated by sgRNAs in A549 cells and was performed by Xu et al[2]. sgRNA #1 matches 227-246 nt, sgRNA #2 matches 209-228 nt, and sgRNA #3 matches 689-708 nt of *Wt PLSCR1*. The sgRNAs likely introduced a short deletion or insertion that does not affect transcription. However, those endogenous mRNA transcripts cannot be translated to functional and detectable *PLSCR1* proteins, as validated by our western blot (below), as well as western blots performed by Xu et al[2]. Therefore, our primers could amplify endogenous *PLSCR1* transcripts upregulated by IAV infection, if 15-244 nt was not disturbed by CRISPR-Cas9 KO. By western blot, we confirmed that only endogenous *PLSCR1* expression is upregulated by IAV infection, and exogenous protein expression of *PLSCR1* plasmids driven by an EF-1α promoter are not upregulated by IAV infection.

Author response image 2.



To avoid confusion, we have removed the original Figure 5A from the manuscript.

(8) In Figure 5C, the loss of anti-flu activity with the H262Y mutant is modest, suggesting the loss of *ifnlr1* transcription is only partly responsible for the susceptibility of *Plscr1* KO cells. The anti-flu activity being independent of scramblase activity resembles the earlier discovery of SARS-CoV-2 (Xu et al., 2024). This could be stated in the results since it is an important point that scramblase activity is dispensable for several major human viruses and shifts the emphasis regarding mechanism. It has been appropriately noted in the discussion.

We appreciated the comments and have acknowledged the consistency of our results with those of Xu et al. under “Both Cell Surface and Nuclear *PLSCR1* Regulates IFN-λ Signaling and

Limits IAV Infection Independent of Its Enzymatic Activity” in the “Results” section.

Reviewer #2 (Recommendations):

(1) The statement that type I interferons are expressed by “almost all cells” is inaccurate (line 61). Type I IFN production is also context-dependent and often restricted to specific cell types upon infection or stimulation.

We apologize for the inaccurate description of the expression pattern of type 1 IFNs and have corrected the restricted cellular sources of type 1 IFNs in the “Introduction”.

(2) The antiviral response is assessed solely through flu M gene expression. Incorporating infectious virus titers (e.g., TCID50 or plaque assay) would provide a more robust and direct measure of antiviral activity.

As requested by the reviewer, we have performed plaque assays on all experiments where flu M gene expression levels were measured (Figure 1G, 5E and 7F, and Supplemental Figure 6E). The plaque assay results are consistent with the flu M gene expressions.

(3) While mRNA expression of interferons is measured, protein levels (e.g., through ELISA) should also be quantified to establish the functional relevance of IFN expression changes.

As requested by the reviewer, we have quantified the protein level of IFN- λ in mouse BAL with ELISA (Figure 2E). The ELISA results are consistent with the mRNA expressions of IFN- λ .

(4) It is unclear whether reduced IFNLR1 expression translates to defective downstream signaling or antiviral responses after IFN- λ treatment in PLSCR1-deficient cells. This is particularly pertinent given the increase in IFN- λ ligand in vivo, which might compensate for receptor downregulation.

We agree with the reviewer that downstream activation of the IFN- λ signaling pathway is a critical aspect of PLSCR1’s proposed regulatory role. To investigate this, we attempted an ISRE-luciferase reporter assay to assess downstream signaling following IFN- λ treatment in PLSCR1 rescue cells. Unfortunately, the experiment encountered unforeseen technical issues. For additional context, please refer to our response to Reviewer #1’s public review #3.

(5) Detailed gating strategies for immune cell subsets are absent and should be included for clarity and reproducibility.

We would like to clarify that the immune cell subsets in BAL fluids were counted manually following cytopspin preparation and Diff-Quik staining (Figure 2B and 7H, and Supplemental Figures 2C, 5D, and 8D), rather than by flow cytometry. We hope this resolves the reviewer’s confusion.

(6) The study does not definitively establish that reduced IFN- λ signaling causes the observed in vivo phenotype. Increased morbidity and mortality in PLSCR1-deficient mice could also stem from elevated TNF- α levels and lung damage, as proinflammatory cytokines and/or enhanced lung damage are known contributors to influenza morbidity and mortality. This point warrants detailed discussions.

We agreed with the reviewer that this study does not guarantee a definitive causality between reduced IFN- λ signaling and increased morbidity of *Plscr1*^{-/-} mice and more experiments are needed to reach the conclusion. We have acknowledged this limitation of

our study in the “Discussion”, as requested by the reviewer. We hope to fully eliminate the confounding elements and definitively establish the proposed causality in future studies.

Reviewer #3 (Public review):

Summary:

Yang et al. have investigated the role of PLSCR1, an antiviral interferon-stimulated gene (ISG), in host protection against IAV infection. Although some antiviral effects of PLSCR1 have been described, its full activity remains incompletely understood.

This study now shows that Plscr1 expression is induced by IAV infection in the respiratory epithelium, and Plscr1 acts to increase Ifn-λ1 expression and enhance IFN-λ signaling possibly through protein-protein interactions on the cell membrane.

Strengths:

The study sheds light on the way Ifnlr1 expression is regulated, an area of research where little is known. The study is extensive and well-performed with relevant genetically modified mouse models and tools.

Weaknesses:

There are some issues that need to be clarified/corrected in the results and figures as presented.

Also, the study does not provide much information about the role of PLSCR1 in the regulation of Ifn-λ1 expression and function in immune cells. This would have been a plus.

We would like to thank the reviewer for the positive feedback and insightful comment regarding the roles of PLSCR1 and IFN-λR1 in immune cells. It is important to note that IFN-λR1 expression is highly restricted in immune cells and is primarily limited to neutrophils and dendritic cells[3]. While dendritic cells were not the focus of this study, we did examine all immune cell subsets in our single cell RNA seq data and performed infection experiments in *Plscr1^{floxStop}/LysM-Cre⁺* mice. We have not observed any significant findings in these populations. On the other hand, we do have some interesting preliminary data suggesting a role for PLSCR1 in regulating Ifn-λr1 expression and function in neutrophils. These findings are discussed in detail in our response to reviewer #3's recommendation #12.

Reviewer #3 (Recommendations):

(1) In Figure 1B, the Plscr1 label should be moved to the y-axis so that readers don't confuse it with the Plscr1-/- mice used in the other figure panels. The fact that WT mice were used should be added in the figure legend.

We apologize for the confusion in the figures. We have moved Plscr1 label to the y-axis in Figure 1B and have mentioned *Wt* mice were used in the figure legend.

(2) In Figure 1C and D, the type of dose leading to the presented data should be added to help the reader. Also, shouldn't statistics be added?

We appreciate the suggestion and have added doses to Figure 1C and 1D. We are confused about the request of adding statistics by the reviewer, as two-way ANOVA tests were used to compare weight losses, and the significance was labeled on the figures.

(3) In Figures 1, F, and G, it is not indicated whether sublethal or lethal dose was used for the IAV infection. This should be very clear in the figure and figure legend.

We apologize for the confusion of infection doses used in the figures. We have added doses to Figure 1F, 1G and 1H.

(4) In Figure 1, the CTCF abbreviation should be explained in the Figure legend.

We have explained CTCF in the figure legend as requested.

(5) In Figure 2B, this is percentages of what?

Figure 2B shows the percentages of each immune cell type within total BAL cells.

(6) In Figures 3A and B, transcriptomes for each condition are from how many mice? Also, what do heatmaps show? Fold induction, differences, etc, and from what? What is compared with what? In addition, is there a discordance between the RNAseq data of Figure 3A and the qPCR data of Fig. 3C in terms of *Ifnlr1* expression?

In Figure 3A and 3C (previously 3B), RNA from the whole lungs of 9 mice per PBS-treated group and 4 mice per IAV-infected group were pooled for transcriptomic analysis. Figure 3A represents a heatmap of differential gene expression, while Figure 3C (previously 3B) represents fold changes in gene expression relative to uninfected controls. In both heatmaps, gene expression values are color-coded from row minimum (blue) to row maximum (red), enabling comparison across groups within each gene (row). The major comparison of interest in these heatmaps is between *Wt* infected mice versus *Plscr1*^{-/-} infected mice. We have added this information to the figure legend.

We also acknowledge the reviewer's observation regarding the discordance between the RNA seq data of Figure 3A and the qPCR data of Figure 3B (previously 3C) for *Ifnlr1* expression. To address this, we have repeated the qRT-PCR experiment with additional samples at 7 dpi. In the updated results, *Wt* mice consistently show significantly higher *Ifn-λr1* expression than *Plscr1*^{-/-} infected mice at both 3 dpi and 7 dpi, consistent with the RNA seq data. However, a time-dependent discrepancy between the RNA-seq and qRT-PCR datasets remains: *Ifn-λr1* expression continues to increase at 7 dpi in the RNA-seq data (Figure 3A), whereas it declines in the qRT-PCR results (Figure 3B). The reason for this discrepancy remains unclear and has been addressed in the Discussion section.

(7) In Figure 3D, have the authors checked whether the *Ifnlr1* antibody they use is indeed specific for *Ifnlr1*? Have they used any blocking peptide for the anti-mouse *Ifn-λr1* polyclonal antibody they are using? Also, in Figure 3E, the marker used for staining should be indicated in the pictures of the lung section.

Unfortunately, a blocking peptide is not available for the anti-mouse *Ifn-λr1* polyclonal antibody used in our study. To assess antibody specificity, we have performed immunofluorescence staining of *Ifn-λr1* on lung tissues from *Ifn-λr1*^{-/-} mice using the same antibody. No signal was detected (Supplemental Figure 5A), supporting the specificity of the antibody for *Ifn-λr1*.

As requested by the reviewer, we have added the marker (*Ifn-λr1*) to the pictures of the lung section in Figure 3E.

(8) In Figure 5, it's better to move each graph's label that stands to the top (e.g. *PLSCR1*, *IFN-λR1* etc) to the y-axis label so that it doesn't get confused with the mouse *-/-* label.

We apologize for the confusion and have moved the top label to the y-axis in Figure 5.

(9) In Figure 6A, it is claimed that the 'two-dimensional UMAP demonstrated that these main lung cell populations (epithelial, endothelial, mesenchymal, and immune) were dynamic over the course of infection.'. This is not clear by the data. The percentage of cells per cluster should be calculated.

As requested by the reviewer, the proportion (Supplemental Figure 6A) and cell count (Supplemental Figure 6B) of each cluster have been calculated and included in "PLSCR1 Expression Is Upregulated in the Ciliated Airway Epithelial Compartment of Mice following Flu Infection" under "Results" section. Together with the two-dimensional UMAP (Figure 6A), these data demonstrate that the main lung cell populations (epithelial, endothelial, mesenchymal, and immune) were dynamic over the course of infection. Following infection, many populations emerged, particularly within the immune cell clusters. At the same time, some clusters were initially depleted and later restored, such as microvascular endothelial cells (cluster 2). Other populations, such as interferon-responsive fibroblasts (cluster 20), showed a dramatic yet transient expansion during acute infection and disappeared after infection resolved.

(10) In Figure 6 B and C, the legend should indicate that these are Violin plots. Also, if AT2 cells don't express *Plscr1*, does that indicate that in these cells *Plscr1* is not needed for *IFN-λR1* expression?

As requested, we have indicated in the legend of Figure 6B and 6C that these are violin plots. *Plscr1* is expressed at low levels in AT2 cells. However, it is unclear whether *Plscr1* is needed for *Ifn-λr1* expression in AT2 cells, and it would be interesting to investigate further.

(11) In lines 302-304, it is stated that 'Among the various epithelial populations, ciliated epithelial cells not only had 303 the highest aggregated expression of *Plscr1*, but also were the only epithelial cell 304 population in which significantly more *Plscr1* was induced in response to IAV infection.'. Which data/ figure support this statement?

Figure 6B shows that among the various epithelial populations, ciliated epithelial cells had the highest aggregated expression of *Plscr1*. To better illustrate this statement, we have rearranged the order of cell clusters from highest to lowest *Plscr1* expression, and added red dots to indicate the mean expression levels for each cluster in Figure 6B.

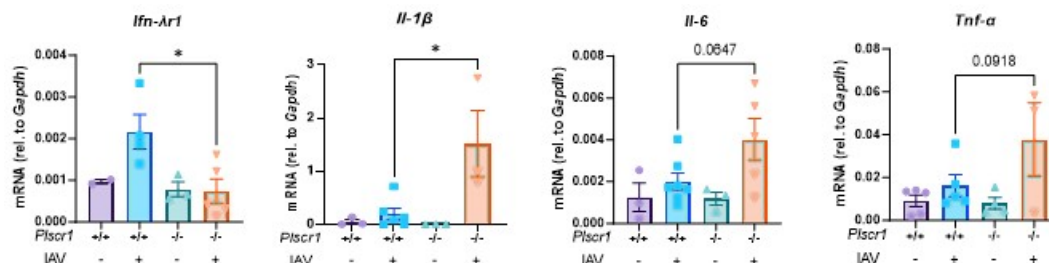
Ciliated epithelial cells also had the most significant increase in *Plscr1* expression ($p < 2.22 \times 10^{-16}$ and $p = 6.7 \times 10^{-5}$) in early IAV infection at 3 dpi (Figure 6C and Supplemental Figure 7A-7K). In comparison, AT1 cells were the only other epithelial cluster to show *Plscr1* upregulation at 3dpi, but to a much less extent ($p = 0.033$, Supplemental Figure 7J). Supplemental Figure 7 was added to better support the statement and the explanation was added to "PLSCR1 Expression Is Upregulated in the Ciliated Airway Epithelial Compartment of Mice following Flu Infection" under "Results" section.

(12) As earlier, if *Plscr1* is not expressed in neutrophils (Figure 6F), does that mean *IFN-λR1* expression does not require *Plscr1* in these cells?

Although *Plscr1* is expressed at lower levels in neutrophils compared to epithelial cells, it is still detectable. In fact, our preliminary data suggest that *IFN-λR1* expression in neutrophils is dependent on *Plscr1*. We have isolated neutrophils from peripheral blood and BAL of IAV-infected *Wt* and *Plscr1*^{-/-} mice using a mouse neutrophil enrichment kit. Quantitative PCR results showed that *Plscr1*^{-/-} neutrophils exhibit significantly lower expression of *Ifn-λr1*, alongside elevated levels of *Il-1β*, *Il-6* and *Tnf-α* in IAV infection (see figures below). These

findings suggest that *Plscr1* may play an anti-inflammatory role in neutrophils by upregulating *Ifn- λ 1*. These data were not included in the current manuscript because they are beyond the scope of current study, but we hope to address the role of *PLSCR1* in regulating *IFN- λ 1* expression and function in neutrophils in a future study.

Author response image 3.



(13) The Figure 7A legend is not well stated. Something like 'Schematic representation of the experimental design of...' should be included. Also, Figure 7J is not referenced in the text.

We apologize for the unclear Figure 7A legend and have changed it to "Schematic representation of the experimental design of ciliated epithelial cell conditional *Plscr1* *KI* mice." Figure 8 (previously Figure 7J) has now been referenced in the text.

(14) In the Methods, more specific information in some parts should be provided. For example, the clones of the antibodies used should be included.

Apart from the 10x technology, the kits used and the type of the Illumina sequencing should be provided. Information on how the QC was performed (threshold for reads/cell, detected genes/per cells, and % of mitochondrial genes etc) should be added.

We apologize for the missing information in the "Methods". We have now provided the clones of the antibodies used, the kit used to generate single-cell transcriptomic libraries, the type of the Illumina sequencing, and the QC performance data.

References

- (1) Rusinova, I., et al., Interferome v2.0: an updated database of annotated interferon-regulated genes. *Nucleic Acids Res*, 2013. 41(Database issue): p. D1040-6.
- (2) Xu, D., et al., *PLSCR1* is a cell-autonomous defence factor against SARS-CoV-2 infection. *Nature*, 2023. 619(7971): p. 819-827.
- (3) Donnelly, R.P., et al., The expanded family of class II cytokines that share the IL-10 receptor-2 (IL-10R2) chain. *J Leukoc Biol*, 2004. 76(2): p. 314-21.

<https://doi.org/10.7554/eLife.104359.2.sa0>